

# **Real options in patenting: Uncertainty and secondary patent markets**

Walter G. Park, American University

Gerard Torres, USPTO

Andrew A. Toole, USPTO and ZEW

Ryan C. Hughes, Addx Corporation and USPTO

USPTO Economic Working Paper No. 2025-2

February 2024

The views expressed are those of the individual authors and do not necessarily reflect official positions of the Office of the Chief Economist or the U. S. Patent and Trademark Office. USPTO Economic Working Papers are preliminary research being shared in a timely manner with the public in order to stimulate discussion, scholarly debate, and critical comment. For more information about the USPTO's Office of the Chief Economist, visit [www.uspto.gov/economics](https://www.uspto.gov/economics)

# Real options in patenting: Uncertainty and secondary patent markets<sup>‡</sup>

Walter G. Park<sup>1</sup>, Gerard Torres<sup>2</sup>, Andrew A. Toole<sup>2,3</sup>, and Ryan C. Hughes<sup>2,4</sup>

<sup>1</sup>American University, Department of Economics

<sup>2</sup>United States Patent and Trademark Office (USPTO)

<sup>3</sup>Leibniz Centre for European Economic Research (ZEW)

<sup>4</sup>Addx Corporation

February 2025

## Abstract

Utilizing a balanced panel of 418 publicly traded companies seeking patent protection in the U.S. from 1999 to 2019, this paper is the first to demonstrate that secondary patent market exposure creates a real option for innovative firms, by allowing for the reversibility of intangible property investments through resale. Consequently, this real option mitigates the negative association between economic uncertainty and patent filings, and can generate a positive association for firms with sufficiently high secondary market exposure, such as those in the technology hardware industry. The reason is that, owing to patent trading, a firm's investments in patent assets are not sunk costs. Through secondary patent markets, firms can utilize patent protection to store value and recoup investments in knowledge when facing economic uncertainty.

**Keywords:** Patents, Innovation, Secondary Markets, Uncertainty, Real Options, Company-level data  
**JEL classification:** O34, O32, E32, E22

---

\*The authors thank Kent Richardson and Erik Oliver for providing data on patent sales. We also thank Michael Andrews, Julian Kolev, Fabio Montobbio, Richard Miller, Nicholas Rada, Ruchi Sharma, Lakhwinder Singh, and Andrea Vezzulli for discussions and Robert Kimble for technical assistance with the project.

<sup>‡</sup>Corresponding author: Gerard Torres - gerard.torres@uspto.gov

# 1 Introduction

Innovation has been surprisingly resilient not only during the recent Covid-19 pandemic, but also during the Great Recession of 2007 and internet bubble of 2000.<sup>1</sup> Certain firms have been able to maintain their innovation activities, if not expand them as evident by their increasing number of patent applications. The objective of this paper is to explain patent application filing behavior during periods of uncertainty. Using insights from real options theory, the paper explains how the resilience of firm-level innovation is related to the availability of a secondary market for patents – a market for trade in ideas – which can turn uncertainty into opportunities. Far from discouraging patent filings, economic uncertainty can in some situations bolster patenting.

This puzzling outcome can be explained by the fact that patents, among other roles, create real options. Holders have the option to reassign or trade their rights to others in the future. Typically under situations of irreversible investment, firms would exercise their option to wait to invest during times of uncertainty. This has the effect of reducing investments. In contrast, when investments are reversible, the negative impact on investment is mitigated. The ability to resell assets later can spur current investments. Since uncertainty is associated with greater variability in returns (or sales or profits), there is the possibility of a higher upside as well as a larger downside. Thus, with innovative firms that have the future option to sell their patent rights, uncertainty can propel firms to seek patent protection for their current innovations. Indeed, our empirical analysis will show that once firms reach some ‘sufficient’ exposure to a secondary market for patents, uncertainty has a positive association with firm patent filings. Otherwise, uncertainty generally reduces incentives to seek patents.

In this paper, we examine the patent filings of 418 public companies seeking patent protection in the U.S. from 1999 to 2019 and consider different measures of uncertainty, such as volatility in sales, stock returns, and cyclical shocks in order to test the robustness of our analysis. Our main finding is consistently observed. Uncertainty typically has a negative association with patenting, as might be expected. However, uncertainty’s negative impact on patent applications is reduced with greater patent resale opportunities, and may even be reversed, controlling for other factors. We show that our results are robust to alternative ways of calculating exposure to a resale market. We also show that there exists some heterogeneity across industries and assess the magnitude of effects.

The key novelties of this paper are that it is the first, to our knowledge, to utilize actual secondary patent market sales data to test the real options theory of patenting, to incorporate the technology fields of a patent to map levels of exposure to secondary patent markets, and to develop more complete counts of patent applications, freer from measurement error, by incorporating

---

<sup>1</sup>See, for example, Buffington, et al. (2021), Davis et al. (2021), and Dinlersoz et al. (2021).

abandoned patent applications and disambiguating assignee names where assignment records are missing for abandoned applications.

Our paper bridges two strands of the literature. The first is the rather large literature on investment and uncertainty, as applied to innovation (intangible) investment. The second is a much smaller literature on secondary markets for patent rights. Combined they help to resolve the puzzle of resilient patent application levels under uncertainty.

The real options framework builds on the seminal work of Dixit and Pindyck (1994). Among other insights, it states that greater uncertainty about revenues may reduce investment in irreversible capital by increasing the value of waiting to make an investment decision. Abel et al. (1996) expand on the framework to show that investment decisions involve the exercise of both reversibility and expandability options. The reversibility option increases the incentives for firms to proceed with an uncertain investment, because purchasing a unit of capital creates the ability to disinvest in the future. On the other hand, the expandability option decreases the incentive for uncertain investments since purchasing a unit of capital extinguishes the ability to purchase that unit in the future. The net effect of uncertainty on current investment is ambiguous, since the two options have offsetting effects on the incentive to invest.

If we apply this option framework to patents, the expandability option has typically been available. For example, under the *Patent Cooperation Treaty (PCT)* system, inventors have the option to delay examination for up to 30 months. This allows inventors more time to determine the commercial potential of their inventions, while at the same time potentially securing priority rights. Thoma (2024) shows that the flexibility to postpone the decision to commit to a technological investment enhances its market value (using the q theory framework). Consistent with real options theory, uncertainty is shown to raise the value of delay.<sup>2</sup> Another study on uncertainty and intangible capital in a real options framework is Bloom and Van Reenen (2002). This study shows that because patents confer exclusive rights, firms can delay the commercialization of their patented innovations under uncertainty. This reduced uncertainty then helps to increase firm productivity by allowing for earlier embodiments of innovation. Neither study addresses the reversibility option of innovation investments nor the role of secondary patent markets.

We extend this real options framework to examine the reversibility and expandability options for patents. While a secondary market for patents exists, it is difficult to obtain transaction data, which partly accounts for the dearth of research on patent secondary markets.<sup>3</sup> Yet, a few prior

---

<sup>2</sup>Thoma (2024) uses the time-invariant dispersion of stock prices (ratio of range of prices to closing price) as the measure of firm-level uncertainty.

<sup>3</sup>Historically, the patent system has helped facilitate trade in technology, as Lamoreaux and Sokoloff (2003)

studies have recognized its role in the innovation ecosystem. Akcigit et al. (2016) point out that up to 20 percent of U.S. patents are traded. These trades allow firms to keep innovations that are aligned with its line of business, to sell those that are technologically distant from its core, and purchase those that are technologically closer to its strategic goals. Resales allow for the improved allocation of ideas across firms and promote economic growth and welfare.

Arora et al. (2022) also discuss how markets for technology enable inventors to sell their inventions to firms that are better able to commercialize them, and show that patents citing scientific articles are more tradable. Serrano (2018) develops a model for estimating the gains resulting from transferring patent rights. Smaller firms, in particular, can transfer their inventions to firms that are more effective at commercialization. Related to this, Ma (2023) finds an interesting trend among U.S. industries whereby production and innovation have separated; that is, some firms specialize in production and others in innovation, particularly smaller firms. Ma (2023) traces this development to the patent trading market, which allows firms to find other ways to monetize their inventions, other than commercializing their inventions and producing the output themselves. However, Arora et al. (2022), Serrano (2018), and Ma (2023) use patent reassignment data to infer the sale of patents. Not all reassignments are sales, however; some might be internal transfers or the updating of information, such as name changes or mergers. Patent reassignments data are reliant on self-disclosure (or voluntary disclosure) by the parties involved.

Love et al. (2018) provide another window into secondary market transactions for patents. This study analyzes the scope of the brokered market for patents using sales transaction data, which they estimate accounts for eighty percent of the entire brokered market. Most of the packages of patents offered for sale were from product-producing technology companies, and the vast majority of inventions pertain to technologies such as computers and telecommunications, with few related to pharmaceuticals and the life sciences. However, the insight into the market for patents discussed in this research has not been incorporated into a real options framework for purposes of explaining patent filing behavior.

It would be useful to mention finance-related research that encircles our core interest. Mezzanotti and Simcoe (2023) analyze the effects of the financial crisis on R&D, finding that cuts to research were more severe for companies with higher short-term debt to cash. Related to this is research on firm borrowing and the ownership of patents. Often, innovators are unable to use tangible capital as collateral for intangible investments and thereby face an external financing

---

document in its study of the U.S. during the late 19th and early 20th century. Patent agents and attorneys played the role of intermediaries to match inventors seeking capital and investors seeking to fund projects, as well as match the sellers and buyers of technologies. These intermediaries lowered the transactions costs of exchange and improved the division of labor between creators and commercializers.

constraint. Thus, borrowing is facilitated if the patents can act as either signals of quality or loan collateral. Milani and Neumann (2022) argue for the signaling effect in their study of how patent assets appeal to external investors to support the R&D of firms experiencing negative internal liquidity shocks. Mann (2018) and Bracht and Czarnitzki (2022) argue for the use of patents as collateral, letting firms borrow more than otherwise. These studies demonstrate the value of having tradeable patent rights.

The rest of the paper proceeds as follows. Section 2 provides a conceptual framework for thinking about the effects of uncertainty and real options on patenting. Section 3 outlines our empirical methodology and Section 4 discusses our data. Section 5 presents our results and Section 6 concludes with some thoughts on the normative implications and future extensions.

## 2 Conceptual Framework

Our study builds on the conceptual framework in Abel et al. (1996). Investment is an intertemporal decision: firms can choose the timing of their investments. Capital may be expanded, contracted, or held constant. When future returns are uncertain, a firm that can purchase capital at a later date possesses a *call option*, while a firm that can resell capital at a later date (assuming reversible investments) possesses a *put option*. Both options affect current incentives to invest, but in opposing directions. On the one hand, higher values of the *call option* create incentives to postpone investment. When faced with uncertainty, firms can expand in the future instead of the present. However, higher values of the *put option* create incentives to invest in the current time period. Only by acquiring capital in the present can the firm exercise an option to sell later.

Consequently, greater future uncertainty tends to have an ambiguous effect on the incentive to invest as it makes both the call and put options more attractive. This helps explain the diverse responses of firms under uncertainty. Economic factors can affect the balancing of the value of the different options.

For example, greater uncertainty may deter R&D investment. high-uncertainty environments may deter firms from making investments. As Czarnitzki and Toole (2011) argue, R&D investment exhibits irreversibility, since much of it is composed of the salaries of R&D personnel and spending on task-specific equipment, which cannot be easily recouped if the research project fails. In response to higher uncertainty, firms may delay or abandon their R&D projects.<sup>4</sup> Czarnitzki and Toole

---

<sup>4</sup>Goel and Ram (2001) find that due to the combination of irreversibility and substantial expandability options, R&D investments react more adversely to uncertainty than non-R&D investments.

(2011) find the ability to patent mitigates the effects of uncertainty on R&D investment, since firms are better able to appropriate the returns to their investment in the presence of rivals or imitators.<sup>5</sup>

In the case of decisions to seek patent protection, the mitigating factor would be the secondary patent market. The secondary market permits patented innovations to be reversible; that is, the owner of a patent has a venue, if needed, for selling patent rights to another party. Thus, given the opportunity to reverse an investment, a firm would either react less negatively to an increase in uncertainty or be spurred to continue to seek patents and avail itself of the ‘put’ options.

We provide an illustration of our hypotheses using the Abel et al. (1996) model. Consider a two-period framework, where in the first period a firm chooses its patent portfolio, called  $x$ . The second period is subject to uncertainty (e.g., economic shock). In the second period, the firm can continue operating with portfolio  $x$ , expand it ( $\Delta x > 0$ ), or contract it ( $\Delta x < 0$ ). Let  $\pi_i$  denote profits in periods  $i = 1, 2$ . The firm’s value is given by:

$$V = \pi_1 + \phi\pi_2^e, \quad (1)$$

where  $\phi < 1$  is the discount factor and superscript  $e$  denotes expected value, given the uncertainty. Let the second period profits depend on a shock,  $\sigma$ , whose cumulative probability distribution is  $F(\sigma)$  with support  $\mathbb{R}$ .

Let  $r$  denote the cost of patenting one invention, broadly speaking to include the costs to apply for a patent and the discounted future costs of maintaining the patent rights.<sup>6</sup> Furthermore, let  $r_L$  be the resale price of a patent and  $r_H$  the purchase price of a patent in period 2, where  $r_L \leq r_H$ . Given these prices, we can identify two critical values of the shock,  $\sigma$ , namely:

$$\frac{\partial \pi_2}{\partial x}(x, \sigma_L) = r_L \quad \text{and} \quad \frac{\partial \pi_2}{\partial x}(x, \sigma_H) = r_H, \quad (2)$$

where the function  $\frac{\partial \pi_2}{\partial x}$  is the marginal return to the patent portfolio in period 2. We assume that the marginal return function is continuous and decreasing in  $x$ , while continuous and increasing in  $\sigma$ .<sup>7</sup> A higher  $r_L$  value creates greater incentive for a firm to acquire patents for sale in period 2. However, a lower  $r_H$  value encourages a firm to delay acquiring patents until period 2. If the marginal return,  $\frac{\partial \pi_2}{\partial x}$ , is below the resale price,  $r_L$ , the firm will divest its patent portfolio until the

---

<sup>5</sup>Czarnitzki and Toole (2007) find that R&D subsidies also help offset the effects of uncertainty on R&D investments.

<sup>6</sup>We can think of a firm budgeting these costs at the time of patenting.

<sup>7</sup>That is,  $\frac{\partial^2 \pi_2}{\partial x^2} < 0$ , which implies diminishing returns to  $x$ , and  $\frac{\partial^2 \pi_2}{\partial x \partial \sigma} > 0$ , which indicates that  $\sigma$  augments the marginal return to patenting.

marginal return rises to equal the resale price. If the marginal return exceeds the purchase price,  $r_H$ , the firm will add to its portfolio until the marginal return falls to equal the purchase price.

Once  $\sigma$  is realized in period 2, the firm might adjust its patent portfolio or keep it the same. The second period profits thus consist of three components: a part attributable to producing with the same patent portfolio,  $x$ , a part due to producing with an expanded patent portfolio less the cost of the additional patents, and a part due to producing with a contracted patent portfolio plus the proceeds from selling patents:

$$\pi_2^e = \int_{\sigma_L}^{\sigma_H} \pi_2(x, \sigma) dF(\sigma) + \int_{\sigma_H}^{\infty} (\pi_2(x', \sigma) - r_H(x' - x)) dF(\sigma) + \int_{-\infty}^{\sigma_L} (\pi_2(x', \sigma) + r_L(x' - x)) dF(\sigma), \quad (3)$$

where  $x'$  is the second period patent portfolio.

Substituting (3) into (1), we can rewrite the firm's value as the sum of three parts:

$$V(x) = R(x) + P(x) - C(x), \quad (4)$$

where

$$R(x) = \pi_1 + \phi \int_{-\infty}^{\infty} \pi_2(x, \sigma) dF(\sigma), \quad (5)$$

$$P(x) = \phi \int_{-\infty}^{\sigma_L} (\pi_2(x', \sigma) - r_L x') - (\pi_2(x, \sigma) - r_L x) dF(\sigma), \quad (6)$$

and

$$C(x) = \phi \int_{\sigma_H}^{\infty} -(\pi_2(x', \sigma) - r_H x') + (\pi_2(x, \sigma) - r_H x) dF(\sigma). \quad (7)$$

$R(x)$  represents the expected present discounted value of profits when there is neither patent resales nor additional patent acquisitions in period 2.  $P(x)$  represents the present discounted value of the put option, which the firm will exercise if the realized shock,  $\sigma$ , falls below  $\sigma_L$ , whereby it sells patents at a price of  $r_L$ . The marginal (discounted) value of the put option is the derivative  $P'(x) = \phi \int_{-\infty}^{\sigma_L} (r_L - \frac{\partial \pi_2}{\partial x}(x, \sigma)) dF(\sigma) \geq 0$ . It is by obtaining a patent in the current period that a firm acquires a put option.  $C(x)$  represents the present discounted value of the call option, which the firm will exercise if  $\sigma$  rises above  $\sigma_H$ , whereby it acquires additional patents at a price of  $r_H$ . The marginal (discounted) value of the call option is  $C'(x) = \phi \int_{\sigma_H}^{\infty} (\frac{\partial \pi_2}{\partial x}(x, \sigma) - r_H) dF(\sigma) \geq 0$ . It is subtracted from firm marginal value, because acquiring a patent in period 1 extinguishes the option to acquire it in period 2.

The value maximizing firm chooses that level of  $x$  to maximize  $V(x) - rx$ . The first-order

condition (F.O.C.) is, using (4):

$$\underbrace{R'(x) + P'(x) - C'(x)}_{q(x)} = r. \quad (8)$$

The left-hand side of (8) altogether represents the marginal value of patents in period 1, which we denote by  $q(x)$ . Holding other factors constant, the higher (lower) the resale price, the greater (lower) the degree of reversibility of patent assets. The marginal value of patents,  $q(x)$ , rises with reversibility, or with the resale price; that is, using  $P'(x)$ , we obtain  $\frac{\partial q(x)}{\partial r_L} = \phi \int_{-\infty}^{\sigma_L} dF(\sigma) = \phi F(\sigma_L) \geq 0$ .

Given that the marginal return functions are decreasing in  $x$ ,  $q(x)$  is decreasing in  $x$ . Figure 1 depicts the optimal  $x$  for different values of the resale price, namely  $\bar{r}_L > \bar{r}_L$ . The optimal current level of the patent portfolio is larger the greater the reversibility option. This is where secondary markets for patents are relevant, for its role in facilitating reversibility. In the absence of such markets, trading is limited, and the resale price, such as it is, is either low or zero.

For a firm facing uncertainty, more valuable call options create incentives to postpone seeking a patent in the current period since the firm has opportunities to invest in the future. More valuable put options motivate a firm to seek patent rights in the current period, since the investment is reversible and the firm can either dampen unexpected losses or profit from resales. An increase in uncertainty as measured by the volatility of returns has an ambiguous effect on investment in patent rights since it raises both the value of the put option and that of the call option -- and both options have opposing influences on the incentive to seek a patent.<sup>8</sup> In the extreme case, where there is no put option (i.e. no reversibility), uncertainty exerts a negative influence on patenting. As put options emerge, or as secondary markets become more available, the negative effects of uncertainty on the decision to apply for patents are mitigated. And with a sufficiently high degree of reversibility, the effect of uncertainty on patent investment decisions may even be positive. Hence the following hypotheses:

**Hypothesis 1 (H1):** *Patent application filings decrease with uncertainty about future returns.*

**Hypothesis 2 (H2):** *Access to secondary patent markets dampen the negative effects of uncertainty on patent application filings.*

---

<sup>8</sup>See Appendix A for technical details.

### 3 Empirical Framework

The unit of analysis is the firm, and the dependent variable of interest is the number of patent applications filed in period  $t$ , measured in quarters. To test the effect of uncertainty on the patent application filings of a firm, conditional on its exposure to the secondary market for patents, we specify the following statistical model for the observed outcomes:

$$P_{it} = \exp(\beta_1 UNC_{it} + \beta_2(UNC_{it} \times ISE_i) + \beta_3 ISE_i + \alpha_0 + \alpha_j + \alpha_t + \alpha_i \bar{P}_i + X'_{it} \delta) \varepsilon_{it}, \quad (9)$$

where the outcome  $P_{it}$  is the number of patent applications at time  $t$  filed by firm  $i$  that were later granted or not granted (for example, abandoned).<sup>9</sup> The independent variable  $UNC_{it}$  is the measure of uncertainty.  $ISE_i$  is an index of exposure to secondary market patent sales, which is explained in more detail in the next section. The vector  $X'_{it}$  includes time-varying firm and industry level controls, time fixed effects ( $\alpha_t$ ), industry fixed effects ( $\alpha_j$ ), and the pre-sample mean of the dependent variable ( $\bar{P}_i$ ), which is discussed below. The time-varying control variables comprise technological push and pull factors, such as R&D stock, sales, industrial concentration, cash flow, credit rating, and firm size as measured by employment. The general constant term is  $\alpha_0$  and the error term is  $\varepsilon_{it}$ , assumed to be mean independent of firm and time, controlling for  $X'_{it}$ :  $E(\varepsilon_{it}|1, UNC_{it}, ISE_i, X'_{it}) = 1$ .

We estimate the exponential model (9) using the Poisson estimator as our patent application data are discrete counts, which include some zero-valued observations. Consistency for standard panel data estimators requires strict exogeneity of the right-hand side (RHS) variables – that is, shocks to patenting at time  $t$  cannot influence future values of the explanatory variables such as R&D (Wooldridge 2010). After testing for and rejecting strict exogeneity, we turned to the pre-sample estimator following Blundell et al. (1999, 2002). This estimator is consistent without imposing strict exogeneity and is implemented by including the pre-sample mean of the dependent variable,  $\bar{P}_i$ , as a separate regressor. We also account for any heteroskedasticity and autocorrelation using HAC robust standard errors following Newey-West (1987).

Our hypotheses (H1 and H2) are that  $\beta_1 < 0$  and  $\beta_2 > 0$ . The threshold level of secondary patent market exposure would be  $ISE^* = \frac{-\beta_1}{\beta_2}$ , such that for firms whose  $ISE_i < ISE^*$ , an uncertainty shock is negatively associated with changes in patenting, and for those whose  $ISE_i > ISE^*$ , an uncertainty shock is positively associated with changes in patenting.<sup>10</sup>

<sup>9</sup>Our empirical specification follows the tradition of the knowledge production approach (Pakes and Griliches, 1984 and Hausman et al., 1984), where patent applications are a function of R&D and other controls.

<sup>10</sup>The threshold  $ISE^*$  is obtained by setting  $\frac{\partial P_{it}}{\partial UNC_{it}} = 0$  in equation (9) and solving for  $ISE$ .

## 4 Data

For our analysis, we constructed a balanced panel with firm-quarter observations. Our data process started with an unbalanced sample of more than 1,800 public and private company names. These companies were among the top patent filers at the USPTO, accounting for about 40% of all U.S. patent application filings between 1990-2021. To obtain financial and other firm-level variables, this group was narrowed to publicly traded companies. We also imposed three other requirements. First, to implement the Poisson pre-sample estimator, the sample companies must have at least one patent application filed in the pre-sample period, which we define as 1990-1998. Second, we restricted our sample to be a balanced panel. This eliminates entry and exit decisions for our sample companies and allowed us to compute Newey-West standard errors.<sup>11</sup> Third, we ended our sample period in 2019 to adjust for incomplete patent application data.<sup>12</sup>

The final sample is a balanced panel of 418 companies that filed for patent protection at least once in the pre-sample (1990-1998) and sample periods (1999-2019), observed at a quarterly frequency. The dependent variable and all explanatory variables are dated by the filing date of the applications; for example, all variables for company  $i$  at year  $t$  and quarter  $q$  are dated at the time the applications were submitted to the USPTO.

### 4.1 Patent Applications

To study a firm’s decision to seek patent protection under uncertainty, we need to accurately measure the dependent variable – the number of applications filed by the firm in each quarter. If we followed prior literature and used either granted patents and/or published applications, many abandoned applications would become part of the error term in equation (9).<sup>13</sup> Having abandoned applications in the error term introduces endogeneity because firm-level explanatory variables influencing the number of applications that are granted such as R&D, employment, sales, and access to external financing also influence which applications the firm chooses to abandon. Therefore, removing abandoned applications from the error term and adding them to our measured dependent variable is both conceptually and empirically appropriate.<sup>14</sup>

---

<sup>11</sup>We programmed our own coding for the Newey-West (1987) standard errors in STATA v. 17.

<sup>12</sup>Patent application data are incomplete after 2019 due to the time between publication and granting/abandonment of applications.

<sup>13</sup>Patent applications are abandoned when the applicant no longer chooses to seek patent protection during the course of patent examination.

<sup>14</sup>To our knowledge prior research has not been able to address this issue due to data limitations.

Our dependent variable,  $P_{it}$ , is the sum of those applications that were eventually granted and those that were eventually abandoned, aggregated by firm and quarter of filing. For granted patents, our main data source is *PatentsView*.<sup>15</sup> For abandoned applications we use USPTO’s internal databases.<sup>16</sup> Figure 2 shows the trend of patent application filings by granted and abandoned status. Most patent applications are eventually granted, but the figure also shows that ignoring abandoned applications would introduce substantial measurement error. For example, in 2006 abandoned applications reached as high as 27% of all patent applications filed in that year.

We went through several additional steps to link our patent applications to filer names and to the firm-level data associated with those filers. We also took into account ownership structures between filing entities, such as subsidiaries, and corporate name changes (e.g. Google to Alphabet). For the names of the filers, this process is easier for granted patents as those typically have an assignee (i.e. owner) on the printed patent. For abandoned applications, the process is described in Appendix B.4.

To sum across patent applications by filer name, the owners’ names must be harmonized so that any text field variations (e.g., IBM and International Business Machines) are properly grouped. *PatentsView* includes disambiguation of assignee names for pre-grant publications and granted patents<sup>17</sup>; however not all applications are public and non-public applications are not included in *PatentsView*. We matched those applications using other methods as described in Appendix B.4.

Next the assignee is matched to its parent company using the *National Establishment Time-Series (NETS)* database supplemented by the *Standard & Poor’s Capital IQ* database. We used a concordance file created when linking *PatentsView* to *NETS* in order to match the multiple variants of assignee/subsidiary names and spellings in *PatentsView* to a common (disambiguated) parent company name in the *NETS* database.<sup>18</sup> The company names in *NETS* were then matched to those in *Capital IQ*,<sup>19</sup> which were used to identify patent applications assigned to foreign companies. The matching involved comparing the assignee (disambiguated) to the list of subsidiaries gathered from *Capital IQ* using the Levenshtein distance method with a threshold of 90 percent. This allowed us to aggregate patent application counts by parent company for firms with headquarters outside of the U.S. With the patent applications finally linked to corporate entities, we use firm-level data from *NETS* and *Capital IQ* for our analysis (see Appendix D for details on data sources).

---

<sup>15</sup>See Appendix B for more details on how patent data are queried from *PatentsView*.

<sup>16</sup>Includes both published and unpublished abandoned applications.

<sup>17</sup>See <https://patentsview.org/disambiguation>.

<sup>18</sup>For more about the process of linking assignees in *PatentsView* to parent companies in *NETS* see <https://www.uspto.gov/sites/default/files/documents/oce-wp-ip-to-nets.pdf>.

<sup>19</sup>We used the relink routine in STATA.

## 4.2 Firm and Industry Variables

We extracted firm-level data from the *Capital IQ* database to use as control variables and to use as inputs for constructing our measures of industry concentration and uncertainty.<sup>20</sup>

*Sales* are revenues earned from the company’s primary activities; they include income, fees, and any premia earned. *Industry concentration* is measured as the Herfindahl-Hirschman Index (HHI):

$$HHI^{jt} = \sum_{i=1}^N s_{ijt}^2, \quad (10)$$

where  $N$  is the number of firms in industry  $j$  and  $s_{ijt}$  the sales share of the  $i^{th}$  firm at time  $t$ . We normalize the  $HHI^{jt}$  so that it varies from zero to one, with higher values indicating greater market concentration.

*R&D spending* consists of expenses in-house or undertaken at outside research facilities (such as universities) for the development of a new product or process, or the upgrading of existing product or service lines.<sup>21</sup> Using the flows of R&D, we constructed *stocks of R&D* using the perpetual inventory method. For each firm:

$$RD_t^{stock} = RD_t^{flow} + (1 - \delta)RD_{t-1}^{stock}, \quad (11)$$

where  $\delta$  is the depreciation rate. The initial stock is given by  $RD_0^{stock} = (\frac{1+g}{g+\delta})RD_0^{flow}$ , where  $g$  is the average growth rate of R&D spending of the firm during the sample period.<sup>22</sup> In the results shown, we used a depreciation rate of 33%. The results are qualitatively the same for variations around  $\delta = 0.33$ .<sup>23</sup> R&D stocks help capture a historical pool of knowledge capital.

We also control for *Cash Flow*, which captures how readily cash can cover items such as liabilities, and can thus gauge firm liquidity. *Cash Flow* is cash from operations, and includes net income, depreciation, amortization, other non-cash sources and changes in working capital. *Full-time employees* are the number of people employed full-time.

For each company, we identified its industry class. We focused mainly on the 4-digit North American Industrial Classification (NAICS) Codes. We also avail ourselves of the broader industry categories provided by *Capital IQ*, such as Information Technology and Hardware, Life Sciences,

---

<sup>20</sup>See <https://www.capitaliq.com/>

<sup>21</sup>Note that R&D would include, where applicable, clinical trials, amortization, and engineering expenses. It would exclude R&D tax credits.

<sup>22</sup>The initial stock is obtained by solving the  $RD_t^{stock}$  difference equation backwards in time.

<sup>23</sup>This depreciation rate is in line with the rate used in Blundell et al. (1999).

Biotechnology, and Pharmaceuticals.

Finally, we control for the financial strength of the firms in our dataset by utilizing available credit data, or *Credit Rating* in our dataset. Our analysis uses data from *Dun & Bradstreet's PAYDEX scores* in *NETS* to accomplish this task. *PAYDEX scores* are available in *NETS* from 1990-2020 and include the minimum and maximum annual scores for firms.<sup>24</sup> We use the annual average of the minimum and maximum *PAYDEX score* for each firm to calculate their quarterly score for each year from 1999-2019. The credit data here are only available for U.S. companies.

### 4.3 Uncertainty

We focus on the definition of uncertainty used by Czarnitzki and Toole (2011) – coefficient of variation of sales – given the conceptual similarities between this study and theirs. Czarnitzki and Toole (2011) focuses on R&D activities in the face of uncertainty, while we analyze the later step in a firm's innovation process – namely, patenting. As robustness checks, we consider three other ways of measuring uncertainty. The first of these is based on Ghosal and Loungani (2000), which derives shocks from the residuals of an autoregressive model. The second is based on stock market volatility, following Bulan (2005). Our third uncertainty measure is based on the business cycle filter method of Baxter and King (1999).

For our primary measure of uncertainty, we make two adjustments to firm-level sales. First, we adjust for inflation by converting nominal sales into real 2012 dollars. Second, we control for industry variations in sales levels and movements. To accomplish this, we normalize firm sales by the industry average sales.

The coefficient of variation is then calculated across quarters ( $t$ ) for each firm ( $i$ ):

$$UNC_{it} = \frac{\sqrt{var(z_{it})}}{mean(z_{it})}. \quad (12)$$

Where  $z_{it}$  is defined as:

$$z_{it} = \frac{sales_{it}}{\frac{1}{N^i} \sum_{i'=1}^{N^i} sales_{i't}}. \quad (13)$$

---

<sup>24</sup>For more about *Dun & Bradstreet's PAYDEX score* see:  
<https://www.dnb.com/resources/what-is-paydex-score.html#infographic>

The numerator in  $z$  is firm sales, while the denominator is the average sales in the firm's respective industry.  $N^\iota$  is the number of firms sampled in industry  $\iota$ . For purposes of calculating the mean and standard deviation of  $z$ , we used past observations up to eight years (as in Czarnitzki and Toole, 2011), depending upon data availability.

As another measure of uncertainty, we employ the method used in Ghosal and Loungani (2000), which fits a fourth order autoregressive model on the profit-sales ratio  $z$ , and uses the standard deviation of the residuals as a measure of uncertainty:

$$z_{it} = \gamma_0 + \gamma_1 z_{it-1} + \gamma_2 z_{it-2} + \gamma_3 z_{it-3} + \gamma_4 z_{it-4} + \gamma_t t + \epsilon_{it}. \quad (14)$$

The standard deviation are computed up to eight years, depending on data availability. We define this alternative measure of uncertainty as:

$$UNC_{it}^A = \sigma(\epsilon_{it}). \quad (15)$$

A third measure of uncertainty is based on the uncertainty in a firm's stock price relative to that of its industry and the market as a whole. Here we follow the methodology of Bulan (2005). Specifically, for each firm, we collected its daily trading stock prices from 1990 - 2021 (where data are available), the values of the S&P 500 market index, and the various S&P 500 industrial indexes (spanning the firms) for each trading day. For each stock market measure (firm, industry, and market) we then calculated the various daily rates of return ( $\rho$ ), excluding non-trading days, which is given by the ratio of the change in the stock price adjusted for dividends ( $S'$ ) to the stock price ( $S$ ):  $\rho = \frac{\Delta S'}{S}$ . Then, we estimated the following two-index regression model, under standard i.i.d. assumptions for returns in each quarter:

$$\rho_{i\tau} = \psi_{it} + \psi_{it}^M \rho_{M\tau} + \psi_{it}^I \rho_{I\tau} + \epsilon_{i\tau}, \quad (16)$$

where  $\rho_{i\tau}$ ,  $\rho_{M\tau}$ , and  $\rho_{I\tau}$  denote the stock market rates of return to firm  $i$ , market  $M$ , and industry  $I$ , respectively.  $\tau$  indexes trading day, where  $\tau = 1, 2, \dots, n$ , and where  $n$  is the number of trading days in each quarter  $t$ . We constructed  $\rho_{I\tau}$  to be orthogonal to  $\rho_{M\tau}$  by regressing the industry returns on market returns and using the residual of the regression as a measure of  $\rho_{I\tau}$ .

It is the residual of this regression from which we derived our measure of the uncertainty in stock market returns. By regressing  $\rho_{i\tau}$  on  $\rho_{M\tau}$  and  $\rho_{I\tau}$ , the above procedure helps to net out the influences of the market (or S&P 500 index) and the industry to which the firm belongs and yield a measure of firm level volatility of returns that is orthogonal to both market and industry fluctuations. The logic here is that merely examining the volatility of the firm's returns does not

inform us of the firm's perceived uncertainty. Uncertainty prevails if the returns are unpredictable beyond what can be observed from fluctuations in the market and industry.<sup>25</sup>

Thus, as a measure of firm level stock market uncertainty, we computed the quarterly standard deviation of the estimated residual ( $\hat{\epsilon}_{it}$ ) – denoted by  $\sigma(\rho_{it})$  – and converted it to quarterly values by multiplying by the square root of the number of trading days per quarter,  $n$ . Typically each quarter contains roughly 60 - 65 trading days. The measure of stock market uncertainty for the firm is thus given by:

$$UNC_{it}^S = \sqrt{n} \sigma(\rho_{it}). \quad (17)$$

Lastly, another approach to measuring economic uncertainty is to examine the volatility of a business cycle indicator. For this purpose, we first use the Baxter and King (1999) filter method to decompose the firm's revenue,  $z$ , into its cyclical and trend components.<sup>26</sup> This is the approach used in Hingley and Park (2017) to study patenting behavior under business cycles. Let  $z_{it} = z_{it}^T + z_{it}^C$ , where  $z_{it}^T$  denotes the trend revenues and the  $z_{it}^C$  its cyclical component. The trend component is positive, but the cyclical component can be positive or negative, being positive when revenues are above trend, and negative when below trend. The natural log of  $z$  then is  $\ln z_{it} = \ln(z_{it}^T + z_{it}^C) = \ln(z_{it}^T(1 + \frac{z_{it}^C}{z_{it}^T})) = \ln z_{it}^T + \ln(1 + \frac{z_{it}^C}{z_{it}^T}) \approx \ln z_{it}^T + \frac{z_{it}^C}{z_{it}^T}$ .<sup>27</sup> The firm-level business shock variable can be defined as  $BK_{it} = \frac{z_{it}^C}{z_{it}^T}$  which is the ratio of the cyclical revenues to trend. The essence of the Baxter-King method is to derive a weighted moving average of the variable concerned,  $z_{it}$ , that is symmetric and centered.<sup>28</sup>

To derive a measure of uncertainty, we compute the volatility in  $BK_{it}$ ; namely the standard deviation of  $BK_{it}$  for the last few years (the equivalent of four years to match the average peak to trough in the U.S. economy).<sup>29</sup> We therefore arrive at:

$$UNC_{it}^B = \sigma(BK_{it}). \quad (18)$$

---

<sup>25</sup>For example, if firm returns move in tandem with those of the market and industry, there is assumed to be limited or no uncertainty.

<sup>26</sup>We have also used the Hodrick and Prescott (1997) filtering method and found qualitatively similar results. The Hodrick Prescott method is a class of high-pass filters in that it removes the trend and retains the high frequency ones. The BK approach is a band-pass filter. It removes frequencies that are deemed both too low and too high to be classified as a business cycle and retains those in-between.

<sup>27</sup>This approximation holds as long as the ratio,  $\frac{z_{it}^C}{z_{it}^T}$ , is not too large.

<sup>28</sup>Our parameter settings for the BK filter were the default ones provided in STATA v.18; that is, the order of the symmetric moving average is 12, and stochastic cycles at periods larger than 32 or smaller than 6 were filtered out.

<sup>29</sup><https://www.nber.org/research/data/us-business-cycle-expansions-and-contractions>.

## 4.4 Index of Secondary Market Exposure

Data on the sale of patents on the secondary market are from *Richardson Oliver Insights*.<sup>30</sup> We analyzed data on packages of patents bought and sold in the U.S. during 2009 - 2021, via brokered patent markets and online platforms, such as *IAM market*.<sup>31</sup> Brokered deals constitute a quasi-public market in that bundles of patents (i.e., packages) are offered to multiple potential buyers. On coverage, the data collected capture approximately 95% of patents placed on this market.<sup>32</sup> Appendix C contains additional information about these packages.

Using the transaction data on patents sold from *Richardson Oliver Insights*, we created a number of indices, which collectively we call the *Index of Secondary Market Exposure (ISE)* to measure firms' exposure to the secondary market. Throughout, we sought to exploit both variation in secondary market activity across technologies (some technologies may have more active trading than others) and variation in technological focus across firms.

To analyze variation in technologies for both the secondary market and firms, we utilize the technology classification according to the Cooperative Patent Classification (*CPC*) system.<sup>33</sup> Patent applications are typically assigned more than one *CPC* symbol, as the underlying inventions can span various technologies. We consider *CPC* symbols at the class level of the *CPC* hierarchy and restrict our attention to all inventional *CPC* symbols. This allows for analysis at both the firm and technology level required for our versions of the *ISE*.

### 4.4.1 Construction of the Index of Secondary Market Exposure

Each of the various forms of our *ISE* have the following structure, where  $i$  indexes firms,  $j$  indexes technologies,  $u$  represents the set of patents or patent applications used in the weighting scheme,

---

<sup>30</sup><https://roipatents.com/>

<sup>31</sup>See Oliver et al. (2019) for an analysis of brokered markets. Private transactions are not included. Love et al. (2018) provide two justifications for their exclusion. First, private transactions are rarely observed, except if they have been disclosed to the SEC. Second, private transactions tend to benchmark their asking prices on brokered transactions, given that the latter are more easily observed. Patents are typically sold in packages, with more than one patent included in the transaction. Typically, only patents are transacted, as pending applications are rarely included in transactions.

<sup>32</sup>This is based on making benchmark comparisons to internal datasets of entities that conduct patent purchases. See Love et al. (2018).

<sup>33</sup>For information about the *CPC* system see: <https://www.cooperativepatentclassification.org>. See USPTO's Manual of Patent Examination Procedure for more details on how patents are classified using the *CPC* system: [905-Cooperative Patent Classification \(uspto.gov\)](#).

and  $w$  corresponds to the specific method used in the weighting scheme to construct a given index:

$$ISE_i^{u,w} = \sum_{j=1}^J Firm\_Share_{ij} \times Weight_j^{u,w}. \quad (19)$$

$Firm\_Share_{ij}$  measures variation in technological focus across firms. The calculation remains the same for all of our versions of the  $ISE$ . This measure is defined as the share of the  $j^{th}$  technological field ( $CPC$  class symbol) in a firm's cumulative patent applications for all of the time periods in our balanced panel dataset, in which there are  $J$  technology categories in total. We adjust for patent applications classified in more than one technological field by dividing the total number of times a  $CPC$  symbol appears in all of a firm's patent applications,  $p_{i,j}$ , by the total number of all  $CPC$  symbols in the firm's patent applications,  $P_{i,J}$ . If a firm has no activity in a given technology field  $j$ , then  $Firm\_Share_{ij} = 0$ . Therefore, we define the following:

$$Firm\_Share_{i,j} = \frac{p_{i,j}}{\sum_{j'} p_{i,j'}} = \frac{p_{i,j}}{P_{i,J}}. \quad (20)$$

We use a weighting scheme,  $Weight_j^{u,w}$ , to measure variation in technologies with secondary market activity. However, we first provide a heuristic overview of the weighting scheme.

Consider Figure 3, which shows a hypothetical distribution of  $CPC$  symbols across patents sold and across patent applications. The large circle on the left represents the total count of all  $CPC$  symbols in patents sold. The smaller inner circles represent the count of specific  $CPC$  symbols among the patents sold, and for simplicity we denote the fields by  $A$ ,  $B$ ,  $C$ , etc. The simple shares of the  $CPC$  symbols in that circle reveal which technologies were most involved in secondary market activity, which is what one weighting scheme relies upon. For example, it appears that technology  $B$  in Figure 3 is the dominant field involved in secondary market transactions. Firms that patent heavily in field  $B$  are then regarded as having good access to a resale market. On the other hand, the right circle in Figure 3 shows that  $B$  is the dominant technology in patent applications. Indeed, if we take ratio of  $B$  in the left circle to  $B$  in the right circle, it would seem that the odds of technology  $B$  being involved in secondary market activity is about one-third, whereas it seems all of the patent applications citing technology  $A$  were involved in secondary market sales. In that regard,  $A$  has far greater odds of being involved in resales than  $B$ . Hence, another way of weighting the importance of a technology in secondary market trading would be to examine its likelihood of being counted in a sold patent.

This is the motivation for why the data for  $Weight_j^{u,w}$  depend upon patent sale transactions from *Richardson Oliver Insights* as well as on patent application counts in our balanced panel

dataset.  $Weight_j^{u,w}$  is constructed along two dimensions: the numerator representing the weight associated with secondary market transaction prices,  $w$ , and the denominator,  $u$ , representing the total set of technology fields in a set of patent applications. Thus:

$$Weight_j^{u,w} = \frac{w}{u}. \quad (21)$$

For the numerator,  $w$ , in  $Weight_j^{u,w}$ , a value of either  $v$  or  $c$  represents the weight given to prices of packages sold in the *Richardson Oliver Insights* dataset. We constructed both a “value-weighted”,  $v$ , and “count-weighted”,  $c$ , version of the numerator  $w$ . The “value-weighted”,  $v$ , numerator attaches higher importance to technologies associated with higher secondary market transaction prices, whereas the “count-weighted”,  $c$ , numerator uses simple counts of *CPC* symbols of the patents transacted on the secondary market.

The denominator,  $u$ , in  $Weight_j^{u,w}$  represents different sets of patent applications in our weighting scheme. We use  $m$  (“marketed”) to represent the total count of *CPC* symbols in patents (granted patent applications) that are sold in the secondary market from the *Richardson Oliver Insights* dataset. Alternatively, we use  $g$  to represent the total count of *CPC* symbols in all (granted or abandoned) patent applications from our balanced panel dataset, whether or not those patent applications go on to be traded in the secondary market.

Expanding on the description above,  $Weight_j^{m,c}$  aims to capture the share of the  $j^{th}$  *CPC* symbol among the set of patents,  $S$ , sold during the twelve year period 2009-2021. Let  $\Omega$  be the total number of all *CPC* symbols occurring in the set of patents  $S$ . Also, let  $\omega_j$  be the number of times *CPC* symbol  $j$  appears in  $S$ . Thus, we arrive at the first version of our weighting schema:

$$Weight_j^{m,c} = \frac{\omega_j}{\Omega}. \quad (22)$$

Next, we describe how we estimate the average market value for a patent sold on the secondary market belonging to a given *CPC* symbol  $j$ . Suppose there are  $B$  bundles of patents, where a bundle is similar to a package from *Richardson Oliver Insights*.<sup>34</sup> Also, suppose  $V^b$  represents the observed price at which bundle  $b$  is transacted and  $K_b$  represents the total number of patents in bundle  $b$ . We assume that the value of a given bundle is distributed equally across its patents. So the value of an individual patent,  $v^p$ , within bundle  $b$  is given by  $v^{b,p} = \frac{V^b}{K_b}$ . Additionally, we assume that that value within a patent is distributed equally among its *CPC* symbols, and use  $t_p$  to represent the number of *CPC* symbols which comprise patent  $p$ . *CPC* symbol  $j$ ’s share of patent  $p$ ’s value within bundle  $b$  is  $v^{b,p,j} = (\frac{V^b}{K_b}) \times (\frac{1}{t_p}) \times \mathbf{1}(p \in j)$ , where  $\mathbf{1}(p \in j)$  is an indicator

---

<sup>34</sup>There is no loss of generality whether a given bundle consists of only a single patent or of many patents.

for whether patent  $p$  belongs to *CPC* symbol  $j$ . We calculate the total secondary market value accruing to *CPC* symbol  $j$ , which we call  $\lambda_j$ , by tallying up  $j$ 's contribution to all patents within a bundle and then aggregating across bundles:

$$\lambda_j = \sum_{b \in B} \sum_{p \in b} v^{b,p,j} = \sum_{b \in B} \frac{V^b}{K_b} \left( \sum_{p \in b} \frac{\mathbf{1}(p \in j)}{t_p} \right). \quad (23)$$

Using the above, we can now formalize our second version of our weighting schema:

$$Weight_j^{m,v} = \frac{\lambda_j}{\left( \sum_{j' \in J} \lambda_{j'} \right)}. \quad (24)$$

For the remaining versions of our *ISE*, we use the set of all patent applications in our balanced panel dataset to form the denominators of  $Weight_j^{u,w}$ . We define  $G_j$  as the total number of patent applications classified to *CPC* symbol  $j$ . This allows us to define the following:

$$Weight_j^{g,c} = \frac{\omega_j}{G_j}, \quad (25)$$

and

$$Weight_j^{g,v} = \frac{\lambda_j}{G_j}. \quad (26)$$

Recalling the definitions above, then, since  $Weight_j^{g,c}$  is a simple ratio of the number of secondary market transactions to the number of patent applications in a given technology field  $j$ , one may think of  $Weight_j^{g,c}$  as a measure of the probability that a patent application involving technology field  $j$  is eventually sold on the secondary market. In contrast, since  $Weight_j^{g,v}$  is calculated as the total secondary market value accruing to technology  $j$  divided by the number of applications involving technology  $j$ , one may consider  $Weight_j^{g,v}$  to be a measure of the average secondary market value per patent application involving technology field  $j$ .

#### 4.4.2 Summary of the Index of Secondary Market Exposure

Finally, we can present all the versions of our *ISE* from the above-mentioned weighting schemes. Our main measure of secondary market exposure is defined as follows:

$$ISE_i^{m,v} = \sum_{j=1}^J Firm\_Share_{ij} \times Weight_j^{m,v}. \quad (27)$$

The main version of the *ISE* attempts to measure a firm’s exposure to the secondary patent market using a value-weighted approach. Technologies included in the most valuable bundles, or packages, are weighted more heavily than those technologies in less valuable bundles. And firms that patent most heavily in those technologies have the most exposure to secondary markets. This is our *preferred* measure of the *ISE* as it relies on the value of historical market transactions to determine the level of exposure to the secondary market.<sup>35</sup>

In addition to a value-weighted approach, we constructed another version of the *ISE* to analyze access to the secondary market based on *CPC* symbol counts. For this version, we set

$$ISE_i^{m,c} = \sum_{j=1}^J Firm\_Share_{ij} \times Weight_j^{m,c}. \quad (28)$$

While  $ISE_i^{m,c}$  utilizes *CPC* symbol counts from actual transactions, it makes the assumption that all *CPC* symbols are of equal value. This assumption is in contrast to our sample data on patent sale transactions and is therefore provided as a comparison to our main measure of the *ISE*.

For further comparison, we constructed another *ISE* using the *CPC* symbols of all the patent applications in our balanced panel dataset as the denominator of the weighting schema. Similar to the main *ISE*, we constructed a value-weighted and count-weighted version defined as follows:

$$ISE_i^{g,v} = \sum_{j=1}^J Firm\_Share_{ij} \times Weight_j^{g,v}, \quad (29)$$

and

$$ISE_i^{g,c} = \sum_{j=1}^J Firm\_Share_{ij} \times Weight_j^{g,c}. \quad (30)$$

These indices may be interpreted as rough measures of the probability that a patent belonging to firm  $i$  is eventually sold on the secondary market or as the expected secondary market value for a given patent application by firm  $i$ , respectively. Both of the above measure access to the secondary market in terms of a technology’s share in our dataset, as opposed to only the transacted patents in the *Richardson Oliver Insights* dataset. It can be thought of as a technology’s propensity to be traded in the market relative to all the other technologies in our sample.

---

<sup>35</sup>Moreover, we stress that this preferred version allows us to exploit information on the prices at which patent packages are sold so as to help us derive an imputed market value of the *CPC* codes embodied in patented inventions and thereby weight the marketability of specific technologies.

### 4.4.3 Analysis of the Index of Secondary Market Exposure

Figure 4 shows a kernel density plot of  $ISE^{m,v}$ , which varies from zero to one for each firm. Exposure to this market is quite skewed across firms. Most firms have little exposure to patent trading. As will be shown, firms in the information technology sector tend to engage the most in patent sales.

Table 1 shows the top twelve *CPC* class symbols classified in the patents sold in the secondary market. In the lead is electric communication technique, symbol H04, followed by computing, calculating, and counting (G06), and basic electric elements (H01). These are technological fields that are typically found in technology hardware. For comparison, the table also shows the top twelve *CPC* class symbols that appear in U.S. licensing transactions over roughly the same time period.<sup>36</sup> Interestingly, half of the top technological fields covered in licensing are among the leading technological fields in the secondary patent markets (see the bolded *CPC* class titles). This concurrence shows these fields to be heavily active in the trading and licensing of patent rights.

Table 2 shows the average exposure to patent secondary markets by industry. Patent sales are prevalent in the communications equipment, audio and video equipment, computers and peripherals, radio, TV, film, and cable broadcasting sectors. Firms that are exposed to secondary patent market sales tend to be in the equipment intensive, high-technology industries.

Table 3 shows some sample statistics for all our main variables, including the *ISE*s and measures of uncertainty. The  $ISE^{m,v}$  displays somewhat more variability than the  $ISE^{m,c}$  in terms of their coefficients of variation. Industrial concentration has the most variability also in terms of the coefficient of variation, and credit rating the least. A summary of the data sources is available in Appendix D.

## 5 Empirical Results

Figure 5 provides a precursor of our results on the relationship among patent application growth, uncertainty, and exposure to the secondary patent market, as measured by the *ISE*. For purposes of comparison, all of the firms in our dataset are categorized by tercile according to their  $ISE^{m,v}$  value. Firms in the first tercile are categorized as “Low Exposure” firms, while firms in the second and third tercile are categorized as “Medium Exposure” and “High Exposure” firms respectively.

---

<sup>36</sup>Data on licensing transactions are from the USPTO’s patent assignment records.

For each tercile, we examine the median patent application growth before and after the maximum uncertainty shock experienced by each firm, which is measured by the largest *change* in sales uncertainty experienced by a firm. Let time  $t$  anchor the quarter at which this shock occurs for a firm. Levels of patent applications are normalized at time  $t - 1$ , the period before the maximal shock. That is, patent application growth is measured as  $P_{it}/\bar{P}_{i0}$ , where the denominator,  $P_{i0} = P_{it-1}$ , is held constant. As shown, the median firm with low secondary patent market exposure experiences a decrease in patent applications following the shock. The median firm with medium exposure also experiences a decline in patent filings, but not to the same extent as the low exposure firm. However, the median high exposure firm experiences a slight increase in applications post-shock. Thus, firms that are highly exposed to the secondary market for patents are spurred to file more patent applications soon after facing a large increase in uncertainty. While this exercise does not control for multiple influences, it motivates us to evaluate how access to secondary markets can mitigate the effect of uncertainty.

## 5.1 Base Results

The main results are presented in Table 4. The evidence shown here supports our hypotheses that uncertainty has a negative association with patent application filings, but that access to secondary markets attenuates the impact of uncertainty. In this table, the measure of uncertainty is the uncertainty in firm's sales and the measure of secondary patent market exposure is based on the value-weighted shares of technological fields in secondary market sales and the firm's share of patenting in those technological fields – namely the index,  $ISE^{m,v}$ .

Column 1 shows estimates of the model with only the uncertainty variable, time and industry fixed effects, as well as the pre-sample mean of the dependent variable. All the other control variables, such as the stock of R&D, sales level, and so forth are absent for purposes of comparison. The results in Column 1 show that uncertainty has a negative association with patent application filings, as the coefficient estimate for sales uncertainty is -2.074, while the elasticity estimate is -0.477.<sup>37</sup> Absent other controls, a 10% increase in uncertainty is associated with a 4.77% decrease in patent application filings.

Column 2 takes into account a firm's exposure to the secondary market for patents by adding the index of secondary market exposure  $ISE^{m,v}$ . The estimated coefficient is significantly positive indicating that firms with higher exposure to secondary markets generally file more patent

---

<sup>37</sup>The elasticity is  $\varepsilon = \frac{\partial P_{it}}{\partial UNC_{it}} \frac{UNC_{it}}{P_{it}} = \beta_1 \times \bar{UNC} = -2.074 \times 0.230 = -0.477$ , where  $\bar{UNC}$  is the mean of sales uncertainty, from Table 3.

applications, all else equal. The patent elasticity with respect to uncertainty is higher, namely -0.588.

Column 3 goes further to determine how a firm's opportunity to sell its patents might diminish the effects of uncertainty by providing a put option. We augment the model with an interaction term,  $UNC \times ISE^{m,v}$ . The coefficient of this term is positive and statistically significant at the 10% level. The positive coefficient indicates that the reduction in patent filings associated with a rise in uncertainty is less, depending on the degree of exposure to secondary patent markets.

In terms of model selection, the likelihood ratio tests conducted ( $-2\ln(\frac{LR}{LU}) \sim \chi^2(\kappa)$ , where  $\kappa$  is the degrees of freedom), show that successively adding the terms  $ISE^{m,v}$  and  $UNC \times ISE^{m,v}$  improves the goodness-of-fit.

Adding a more complete set of control variables in columns 4-6 preserves the main qualitative findings in columns 1-3 but strengthens the finding that secondary patent markets help alleviate the negative impact of uncertainty on patent application filings. In column (4), the uncertainty variable appears alone without the index of secondary patent market exposure. In column (5), the index is added as another independent variable, and in column (6), the interaction between the index and uncertainty is included. Progressively, these results show that uncertainty has a negative association with patent filings while exposure to secondary markets can boost filings. Inclusion of the interaction term ( $UNC \times ISE^{m,v}$ ), which again has a significantly positive estimated coefficient, shows that exposure to secondary patent markets offsets the effects of uncertainty.

Focusing on the full estimated model in column (6), we find that the coefficient estimates for uncertainty of -2.879 and for the interaction term of 5.507, with both being significant at the 1% level, yield an estimated elasticity of -0.22.<sup>38</sup> A 10% increase in uncertainty is associated with a 5.67% decrease in patent application filings, on average. Ignoring this exposure to secondary patent markets and basing only on the coefficient of  $UNC$  would have suggest a 28.79% decrease in applications. This shows the mitigating effects of exposure to the secondary market. The threshold value  $ISE^* = \frac{-\beta_1}{\beta_2} = \frac{2.879}{5.507} = 0.52$ . For the complete sample, most firms are below this threshold, so that most companies' patent filings decrease on balance when uncertainty rises. (However, when we examine the data by industry, we find greater heterogeneity in the estimated threshold values.)

The likelihood ratio tests show strong support for the hypothesis that secondary patent market exposure matters for the firm's decision to apply for a patent. They support the idea that real op-

---

<sup>38</sup>The derivation is as follows:  $\frac{\partial \ln P_{it}}{\partial UNC_{it}} = -2.879 + 5.507 \times ISE^{m,v}$ . Thus, the elasticity is  $\varepsilon = \frac{\partial P_{it}}{\partial UNC_{it}} \frac{UNC_{it}}{P_{it}} = (-2.879 + 5.507 \times ISE^{m,v}) \times UNC = -0.567$ , where the mean sales uncertainty  $UNC = 0.230$  and the mean  $ISE^{m,v} = 0.075$ , from Table 3, which provides the descriptive statistics.

tion thinking is present in their managerial decision-making. Both the exposure to such secondary markets and the interaction of the exposure and uncertainty add value in terms of explanatory power and model fit.

As for the control variables, R&D stock, sales, and cash flow, have positive and statistically significant associations with patent application filings. Employment has a positive, but not statistically significant influence at conventional levels. Credit rating is significant when the index of secondary market exposure is not controlled for. Industrial concentration (HHI) has a significant, negative association. The latter indicates that greater levels of application filings arise in less concentrated and more competitive industries, controlling for other factors. Overall, the model with a full set of RHS variables explains almost 70% of the variations in the data, according to the Pseudo- $R^2$ . Without the full set, the model explains around 60% (see columns 1-3 of Table 4).

## 5.2 Alternative Indexes of Secondary Patent Market Exposure

In this section, we check the sensitivity of our results to how we measure a firm’s exposure to secondary patent markets by considering different formulations of the index of secondary market exposure.

The  $ISE^{m,v}$  index was calculated using the imputed value of *CPC* symbols so that a firm’s exposure to secondary patent markets was based on its value-weighted shares in transactions. The market values of *CPC* symbols are rough estimates. First, as a robustness check, we establish that the results are similar if we had not value-weighted a firm’s share in secondary patent market transactions, but simply calculated the share in terms of counts (of *CPC* symbols involved in secondary trading).

Columns 1-2 in Table 5 therefore repeat the analyses using the  $ISE^{m,c}$  index. The results mirror those in columns 5 and 6 of Table 4. The measured quantitative estimates are very similar as are the significance levels. However, the main difference between the two indexes  $ISE^{m,c}$  and  $ISE^{m,v}$  is that the latter is more positively skewed. There is somewhat more density in the lower ranges of  $ISE^{m,c}$ .

The  $ISE^{m,c}$  essentially measures how exposed a firm is to the secondary market by examining its distribution of patents by technological fields, and then examining whether those technological fields show up in secondary patent market sales (by count or frequency). The latter is one way of determining which technological field is more tradable than others. An alternative way of determining it is to calculate the likelihood that a patent belonging to a given a technology field

will be involved in secondary market sales. This requires comparing the sample of patents sold to all the patent applications in our sample. Hence, this likelihood is captured by the ratio of patents sold in a specific *CPC* symbol to all patent applications in that specific technology.

Columns 3-4 of Table 5 display the results of the model using the modified index, namely the measure we derived earlier,  $ISE^{g,c}$ . Again, in this version, the representation of a technology is based on the ratio of counts of *CPC* symbols in sold patents to the counts of *CPC* symbols in all patent applications in our sample. The estimates support our hypotheses that  $\hat{\beta}_1 < 0$  and  $\hat{\beta}_2 > 0$ , where both coefficient estimates are significant at the 1% level. Secondary patent market exposure is shown to diminish the effects of uncertainty on filings. Lastly, columns 5-6 of Table 5 show the results of using the  $ISE^{g,v}$  version, where a firm's technological fields are weighted by the average secondary market value per patent application. Once more, our hypotheses hold strongly. LR ratio tests again show significant marginal gain from modeling secondary patent market exposure.

### 5.3 Alternative Measures of Uncertainty

We return to using the original index of secondary patent market exposure,  $ISE^{m,v}$ . Our next robustness check is to examine whether our hypotheses hold under alternative measures of uncertainty, as described in Section 4.3.

Table 6 presents the results. In all cases, we observe the same pattern for our key coefficients:  $\hat{\beta}_1 < 0$  and  $\hat{\beta}_2 > 0$ . First, in columns 1-3, sales uncertainty is measured not as a coefficient of variation but as the standard deviation of the residuals from an auto-regressive model of sales of order four plus a trend term. We denote this measure by  $UNC^A$ . This variable on its own, or with  $ISE^{m,v}$  or its interaction with  $ISE^{m,v}$ , has a statistically significant negative coefficient at the one percent level of significance. In column 3, its coefficient value of -0.017 and the coefficient value of the interaction term of 0.044 indicates that the threshold value of  $ISE^{m,v}$  is 0.39. Firms with a threshold value above this are expected to have a positive association between patent filings and uncertainty; that is, are spurred to exploit the put option to file more, while those firms below the threshold would postpone patent filings under uncertainty.

For the results in columns 4-6, we use uncertainty in the firm's stock market rate of return (after netting out industry-wide and market-wide rates of return). The stock market uncertainty measure (denoted by  $UNC^S$ ) also provides results that are similar to earlier results. Uncertainty, unaccompanied by the secondary patent market variable, has a negative association with patenting at the 10% level of significance. When the secondary patent market index and its interaction with uncertainty are included, stock market uncertainty is significantly negative at the 1% level. The

interaction between stock market uncertainty and the index of secondary market exposure has a statistically significant positive coefficient. The estimated threshold for  $ISE^{m,v}$  is 0.13 (using estimates in column 6). For firms with exposure greater than this critical value, uncertainty has a net positive association with patent applications. Just under a fifth of firms have  $ISE$  levels above 0.13.

Lastly, we present the results of using uncertainty in business cycles – namely, the volatility of a business cycle measure (denoted by  $UNC^B$ ) that is derived from the Baxter-King trend-cycle decomposition method. As discussed earlier, from this decomposition, we computed the volatility in the ratio of a firm’s cyclical sales to its sales trend. It should be noted that the key difference between our business uncertainty measure and business cycles is that our uncertainty measure is another way of capturing the volatility of sales, whereas the business cycle decomposition captures episodes of recessionary pressures and booms.<sup>39</sup> As shown in columns 7-9, business cycle uncertainty has a negative association with patent filings, while the interaction between that type of uncertainty and the exposure to secondary patent markets has a strongly significant positive association. The estimated threshold for  $ISE^{m,v}$  is 0.32 (using estimates in column 9).

All three measures support our hypotheses. Whichever way uncertainty is measured, exposure to secondary patent markets offers firms the option to sell their patent assets ex post. This helps to dampen the effects of an uncertain market environment. While the qualitative effects are similar, quantitatively the different ways of measuring uncertainty do yield different estimates of the critical threshold. They all show that only a minority of firms actually increase their patent filings during high uncertainty. Most respond by postponing or decreasing their filings, but the impact of uncertainty is nevertheless lessened by the availability of secondary patent markets. As we will see in the next section, the critical values of  $ISE$  vary by industry. For that reason, we defer more detailed analyses of companies that are above the threshold to the next section.

The control variables largely show similar qualitative (and quantitative) results as found in Tables 4 and 5. The level of sales, the stock of R&D, cash flow, and credit ratings have positive associations with filing patent applications, while industry concentration (HHI) has a negative association and employment size an insignificant association.

---

<sup>39</sup>For example, when the cycle-trend ratio is negative, a firm’s sales is below trend and the firm is experiencing a downturn. However, when the same ratio is positive, a firm is experiencing an upswing.

## 5.4 Sector Analysis

In this section, we analyze sub-samples of the data using industry groups from *S&P Capital IQ*. We ran separate regressions for each industry group. To conserve space, we report in Table 7 the coefficient estimates for the key variables of interest:  $UNC$ ,  $ISE^{m,v}$ , and  $UNC \times ISE^{m,v}$ . Our hypotheses ( $\beta_1 < 0$  and  $\beta_2 > 0$ ) hold in the following sectors: Consumer Durables, Information Technology, Materials, Retail & Marketing, and Utilities & Energy. For the rest,  $\hat{\beta}_2$  is statistically insignificant.

For each industry group, Table 7 also displays the  $ISE$  threshold value, the number of observations and the number of firms in each industry, the number and percentage of firms with an  $ISE^{m,v}$  value above the industry  $ISE$  threshold value, as well as the median values of the  $ISE^{m,v}$  for the respective industry group.

As Table 7 shows, the threshold values of  $ISE^{m,v}$  are industry-specific; for example, it is 0.238 for Consumer Durables, where only 3 out of 35 firms are above it, and threshold is 0.118 for Utilities & Energy, where 10 out of 27 firms are above it. However, the information technology industry stands out, as almost half the firms (i.e., 48%) have exposure levels to the secondary patent market that are above the  $ISE$  threshold. This is not surprising, given both the number of firms in this industry group and the technological distribution of secondary market patent sales shown in Table 1. Many of the patents sold on the secondary market relate to information technology. Thus, firms in the information technology industry group are rather highly engaged in secondary patent market trading, and their patent filings are expected to react positively to uncertainty shocks.<sup>40</sup>

Prominent innovations that are largely absent from secondary patent markets are drugs and biotechnological innovations. This can help explain why the patent filings of firms in the pharmaceutical sector react only negatively to uncertainty; that is,  $\hat{\beta}_1$  is negative and statistically significant at the 1% level but  $\hat{\beta}_2$  is not significant at conventional levels. Firms in the pharmaceutical sector have very low levels of exposure to the patent secondary market, as shown in Table 2 (see NAICS code 3254). Innovations in drugs tend to have longer product cycles and pharmaceutical products tend to be discrete products and are not as modular in nature as IT products, rendering them less amenable to secondary market trading. These findings are consistent with what Love et al. (2016) observed about the brokered market, which is that IT companies are

---

<sup>40</sup>Note that the coefficient of  $ISE^{m,v}$  is estimated to be negative for the information technology sector; that is, holding all else constant, firms that are more exposed to secondary patent markets have fewer patent applications on average. This is because of the presence of a number of young firms in this sector (or start-ups), where firm age is based on year of founding. Younger firms in the IT sector tend to engage heavily in patent trading, reassignments, and licensing, and they also tend to patent less than the larger, more established companies.

highly active while pharmaceutical firms are not.

Overall, for most of the firms in our sample, we observe a negative association between patent applications and uncertainty. Fewer than 20% of companies are above the critical thresholds. Specifically, 77 out of 418 firms are above the threshold as shown in Table 7.

## 5.5 Quantitative Effects of Uncertainty

Table 8 provides some insight on the quantitative significance of the exposure to secondary markets. Using the results in Table 7, namely applying the coefficient estimates by sector to the firms in that sector, we compute the *elasticity* of patent filings with respect to uncertainty and determine the rough estimate of how much patent filings would change under a half standard deviation increase in uncertainty. From equation (9), the elasticity ( $\epsilon$ ) is computed as:

$$\frac{\partial P}{P} \frac{\overline{UNC}}{\partial \overline{UNC}} = (\hat{\beta}_1 + \hat{\beta}_2 \overline{ISE}) \times \overline{UNC}, \quad (31)$$

where  $\overline{UNC}$  is the mean uncertainty, based on the coefficient of variation in sales, and  $\overline{ISE}$  is the mean index of secondary market exposure,  $ISE^{m,v}$ , as based on the value weighted shares of *CPC* symbols in secondary market sales.

The expected change in patent applications,  $\Delta P$ , is based on the percentage increase or decrease over the sample quarterly mean filings of patents.<sup>41</sup> The upper part of Table 8 shows aggregate changes in filings as well as filings broken by tercile of  $ISE^{m,v}$ . The rest of the table shows the changes by select firms.

The overall change in filings depends on how much patenting was done and on how much access a firm has to the secondary patent market. Since firms with low exposure to secondary markets also file fewer applications on average, their decreases in patent filings are relatively small. Among companies with significant exposure, high exposure firms file more applications than medium exposure firms on average. However, the decline in patent filings owing to a rise in uncertainty is smaller for the high exposure firms because of their resale options. The last column in Table 8 shows that on average (for the pooled sample), a half standard deviation increase in uncertainty is associated with a 3 percent decrease in mean patent applications. The percentage decrease is highest for the low secondary patent market exposure group and least for the high exposure group.

For all firms in the low and medium exposure groups, patent filings are expected to fall. For

---

<sup>41</sup>  $\Delta P = \frac{\Delta UNC}{UNC} \times (\frac{\epsilon}{100}) \times \bar{P}$ , where  $\Delta UNC$  is half the standard deviation of  $UNC$  during the sample period.

some firms in the high exposure group, patent filings are also expected to fall (see the outcome for IBM). But for firms with sufficiently high exposure to secondary markets, filings are expected to rise (see the outcome for Apple, Qualcomm, and Cisco). In terms of magnitude, a one-half standard deviation increase in uncertainty can result in about one fewer filing per quarter for a company like *Goodyear*, or a 4.5% decrease. For a pharmaceutical company like Pfizer, a half-standard deviation increase in its uncertainty is associated also with about one fewer filing per quarter, or a 1.4% decrease in its mean quarterly patent filing; for *Merck*, the same rise in uncertainty is associated with a decrease in filings per quarter by about 2, or a 3.2% drop. For *Microsoft*, it is associated with a rise of about one more filing per quarter, or a 0.2% increase.

A comparison of companies across secondary market exposure levels helps illustrate the value of access to such a market. A few firms in the higher exposure groups experience a rise in patent filings amid a rise in uncertainty. If we compare *Boeing* and *Motorola*, both have similar levels of patent filings, while *Motorola* has about three times the exposure to secondary patent resale markets. It also has experienced much higher levels of uncertainty during the sample period, yet under the same kind of uncertainty increase, *Boeing's* filings are expected to fall by 9.36 filings per quarter (or by 4.4%), while *Motorola's* is expected to rise by 21.42 (or by almost 10%). *NVIDIA* files roughly the same number of applications per quarter as does *Thermo Fisher Scientific*. Both face the same percentage increase in uncertainty under a half standard deviation increase in uncertainty. Yet, *NVIDIA's* quarterly filings would rise by 1.68 (or 3.2%) while those of *Thermo Fisher Scientific* would fall by 1.6 (or 3.8%).

## 6 Conclusion

This paper offers a new perspective on how economic uncertainty and secondary markets influence the decision to file for patent protection. Using a real options framework, we show that the number of applications filed for patent protection generally falls as uncertainty increases, but that exposure to secondary markets can weaken this negative effect. Through secondary markets, patents can be sold to recoup at least part of the initial capital investment, thereby creating “reversibility options.” We find that for firms with a sufficiently high exposure to secondary markets – above a critical threshold level – the net effect of uncertainty on patent application filings turns positive (controlling for other influences).

This paper developed four variants of an index of secondary patent market exposure and estimated the threshold index values for our preferred index across industry groups. At these critical values, just under 20% of the firms in the sample would be predicted to increase their patent

application filings in response to increases in uncertainty. Thus, they are overall a minority of companies. These companies tend to be in the technology hardware or software sectors, where innovation inputs are tradable. These kinds of industries have been characterized in previous work as exhibiting sequential, cumulative innovations, or as developing modular technologies, where cross-licensing often occurs to facilitate or enable the practice of inventions.<sup>42</sup> These features should help make it possible to trade and reuse, or re-purpose, technologies in other firms.

Our results are robust to alternative measures of uncertainty, such as uncertainty in stock returns, or using other methods of deriving uncertainty such as the use of an AR model to generate shocks or the use of business cycle filtering methods to split movements in sales between its trend and cyclical components.

As mentioned, how firms respond to uncertainty and secondary markets is heterogeneous across industries. Reactions vary depending on how heavily their technologies are involved in secondary patent market trading. For example, the *CPC* symbols typically assigned in pharmaceutical patents are not among those in traded patents. Hence, we found the coefficient of the interaction term (between uncertainty and secondary market exposure) to be statistically insignificant for the pharmaceutical and biotech industries, indicating that exposure to secondary patent markets did not attenuate the effects of uncertainty shocks on pharmaceutical patent application filings.

The overall takeaway in this study is that the reversibility of investments in patent assets, via the outright sales of patent rights, is an important determinant of patent filing behavior that has not to-date been demonstrated.

Our paper is novel in being the first to utilize observed secondary patent market data on the quantity and prices of patent packages sold. We were able to measure the level of exposure that a firm has to the secondary patent market by taking advantage of the information revealed about the composition of the *CPC* codes in the packages sold; in other words, information about the technological fields that are the objects of trading in secondary patent markets. We then connected that information back to a firm’s technological specializations in their patent applications and thereby inferred the degree of exposure or likelihood of exposure to secondary patent markets. Another novelty of this paper is that it carefully derives measures of patent applications by using data on abandoned and granted patents, thereby removing the possibility of endogeneity owing to abandoned patents being in the error term.

The paper has implications for firm strategy and public policy. On firm strategy, companies can utilize the put options or call options associated with patent filings. Companies that are

---

<sup>42</sup>See, for example, Hall and Ziedonis (2001) and Ivus et al. (2017).

capable of accessing secondary patent markets need not alter their filings radically in the face of an uncertainty shock. However, firms with very little to no exposure to secondary patent markets can only utilize the call option – that is, postpone their filings – in the face of market uncertainty. On public policy, there may be a case for policy makers to support secondary patent markets for the diversification and risk sharing benefits they can provide. Lessening the adverse impacts of uncertainty can foster resilience in innovation activity, particularly in emerging markets. As argued in Akcigit et al. (2016), an expansion in the market for ideas and improved intermediation have the potential to increase productive efficiency.

For future extensions, it would be useful to carry out this study in other regions, such as Europe or Asia, and examine whether there are options to participate in secondary patent markets. Second, a deeper dive into the sources of reversibility or trade-ability of patent rights would be illuminating. What makes a technological invention sector-specific or sector-general? Third, we did not explore how patent rights get sold. For example, when a firm offers its invention for sale, prices of asset packages should adjust to ensure an equilibrium between demand and supply. If a seller faces revenue uncertainty, so might a buyer. If many companies experience similar degrees of uncertainty, the resale market may be depressed. This should influence the ease with which firms can exercise their put options to sell patent rights. Thus, sale prices will depend on general industrial and macroeconomic conditions. Exploring these aspects calls for a more integrated micro and macro analyses.

## References

- [1] Abel, Andrew, Dixit, Avinash, Eberly, Janice, and Pindyck, Robert. 1996. “Options, the Value of Capital, and Investment.” *Quarterly Journal of Economics* 111(3): 753-777.
- [2] Akcigit, Ufuk, Celik, Murat A., and Greenwood, Jeremy. 2016. “Buy, keep, or sell: Economic growth and the market for ideas.” *Econometrica*, 84(3): 943-984.
- [3] Arora, Ashish, Belenzon, Sharon, and Suh, Jungkyu. 2022. “Science and the market for technology.” *Management Science*, 68(10): 7176-7201.
- [4] Baxter, Marianne and King, Robert. 1999. “Measuring Business Cycles: Approximate Band-pass Filters for Economic Time Series.” *Review of Economics and Statistics* 81(4): 575-593.
- [5] Bloom, Nicholas and Van Reenan, John. 2002. “Patents, Real Options, and Firm Performance.” *Economic Journal* 112: C97-C116.
- [6] Blundell, Richard, Griffith, Rachel, and Van Reenan, John. 1999. “Market Share, Market Value and Innovation in a Panel of British Manufacturing Firms.” *Review of Economic Studies* 66: 529-554.
- [7] Blundell, Richard, Griffith, Rachel, and Windmeijer, Frank. 2002. “Individual Effects and Dynamics in Count Data Models.” *Journal of Econometrics* 108: 113-131.
- [8] Bracht, Felix and Czarnitzki, Dirk. 2022. “Patent Collateral and Access to Debt.” ZEW Discussion Paper No. 22-033.
- [9] Buffington, Catherine, Chapman, Daniel, Dinlersoz, Emin, Foster, Lucia, and Haltiwanger, John. 2021. “High Frequency Business Dynamics in the United States During the COVID-19 Pandemic,” U.S. Census Bureau, CES working paper 21-06.
- [10] Bulan, Laarni, T. 2005. “Real Options, Irreversible Investment and Firm Uncertainty.” *Review of Financial Economics* 14: 255-279.
- [11] Czarnitzki, Dirk and Toole, Andrew A. 2007. “Business R&D and the Interplay of R&D Subsidies and Product Market Uncertainty.” *Review of Industrial Organization* 31(3): 169-181.
- [12] Czarnitzki, Dirk and Toole, Andrew A. 2011. “Patent Protection, Market Uncertainty, and R&D Investment.” *Review of Economics and Statistics* 93: 147-159.
- [13] Davis, J., Holden, Diethorn, Marschke, Gerald, and Wang, Andrew. 2021. “STEM Employment Resiliency During Recessions: Evidence from the COVID-19 Pandemic,” NBER Working Paper No. 29568.
- [14] Dinlersoz, Emin, Dunne, Timothy, Haltiwanger, John, and Penciakova, Veronika. 2021. “Business Formation: A Tale of Two Recessions,” U.S. Census Bureau, Center for Economic Studies Paper No. CES-WP-21-01.

- [15] Dixit, Avinash and Pindyck, Robert. 1994. *Investment Under Uncertainty*. Princeton University Press.
- [16] Ghosal, Vivek and Loungani, Prakash. 2000. "The Differential Impact of Uncertainty on Investment in Small and Large Businesses." *Review of Economics and Statistics* 82(2): 338-349.
- [17] Goel, Rajeev and Ram, Rati. 2001. "Irreversibility of R&D investment and the adverse effect of uncertainty: Evidence from the OECD countries." *Economics Letters* 71: 287-291.
- [18] Hausman, Jerry, Hall, Bronwyn and Griliches, Zvi. 1984. "Econometric Models for Count Data with an Application to the Patents R and D Relationship." *Econometrica* 52: 909-938.
- [19] Hall, Bronwyn and Ziedonis, Rosemarie. 2001. "The patent paradox revisited: An empirical study of patenting in the US semiconductor industry, 1979-1995." *RAND Journal of Economics* 32(1): 101-128.
- [20] Hingley, Peter and Park, Walter G. 2017. "Do Business Cycles Affect Patenting? Evidence from European Patent Office Filings." *Technological Forecasting and Social Change*, 116: 76-86.
- [21] Hodrick, Robert and Prescott, Edward C. 1997. "Postwar U.S. Business Cycles: An Empirical Investigation." *Journal of Money, Credit, and Banking*, 29(1): 1-16
- [22] Ivus, Olena, Park, Walter, and Saggi, Kamal. 2017. "Patent Protection and the Composition of Multinational Activity: Evidence from U.S. Multinational Firms." *Journal of International Business Studies* 48(7): 808-836.
- [23] Lamoreaux, Naomi and Sokoloff, Kenneth. 2003. "Intermediaries in the U.S. Market for Technology." in S. Engerman, P. Hoffman, J-L Rosenthal, and K. Sokoloff, eds., *Finance, Intermediaries, and Economic Development*. Cambridge University Press: 209-246.
- [24] Love, Brian, Richardson, Kent, Oliver, Erik, and Costa, Michael. 2018. "An Empirical Look at the 'Brokered' Market for Patents." *Missouri Law Review* 83: 359-407.
- [25] Ma, Yueyuan. 2023. "Specialization in a Knowledge Economy." SSRN Working Paper No. 4052990.
- [26] Mann, William. 2018. "Creditor rights and innovation: Evidence from patent collateral." *Journal of Financial Economics*, 130: 25-47.
- [27] Mezzanotti, Filippo and Simcoe, Timothy. 2023. "Research and/or Development? Financial Frictions and Innovation Investment." NBER Working Paper No. 31521.
- [28] Milani, Sahar and Neumann, Rebecca. 2022. "R&D, Patents, and Financing Constraints of the Top Global Innovative Firms." *Journal of Economic Behavior and Organization*, 196: 546-567.
- [29] Newey, Whitney and West, Kenneth. 1987. "A Simple, Positive Semi-definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix." *Econometrica*, 55(3): 703-708.

- [30] Oliver, Erik, Richardson, Kent, and Costa, Michael. 2019. “Tracking Buyer and Seller Behavior: 2018 Patent Market Report.” <https://www.ipwatchdog.com/2019/05/26/>
- [31] Pakes, Ariel and Griliches, Zvi. 1984. “Patents and R&D at the Firm Level: A First Look.” in Z. Griliches, ed., *R&D, Patents, and Productivity*. Chicago: University of Chicago Press: 55-72.
- [32] Serrano, Carlos. 2018. “Estimating the Gains from Trade in the Market for Patent Rights.” *International Economic Review*, 59(4): 1877-1904.
- [33] Thoma, Grid. 2024. Market valuation of international patenting using a real option approach: the moderating role of uncertainty. *Economics of Innovation and New Technology*, 1–22.
- [34] Wooldridge, Jeffery M. 2010. *Econometric Analysis of Cross Section and Panel Data*, MIT Press.

## A Appendix: Uncertainty and Put, Call Options

We use our conceptual model to elaborate why uncertainty in returns has an ambiguous effect on the incentive to seek patent protection. Denote  $\frac{\partial \pi_2}{\partial x}(x, \sigma)$  by  $z$ . The distribution of  $z$  is shaped by the distribution of shocks,  $\sigma$ . Let  $G(z)$  be the cumulative distribution function induced by  $F(\sigma)$ .

Recall that the discounted marginal value of the put option is  $P'(x) = \phi \int_{-\infty}^{\sigma_L} (r_L - z) dF(\sigma)$ . With a change of variables, we get:

$$P'(x) = \phi \int_0^{r_L} (r_L - z) dG(z) \quad (\text{A1})$$

where  $\sigma \in (-\infty, \sigma_L]$  mapped into  $r_L \in [0, r_L]$ . Integrating by parts (i.e.,  $\int u dv = uv - \int v du$ , where  $u = (r_L - z)$ ,  $du = -dz$ ,  $dv = dG(z)$ , and  $v = G(z)$ ) gives us:

$$P'(x) = \phi \left( (r_L - z) G(z) \Big|_0^{r_L} + \int_0^{r_L} G(z) dz \right) \quad (\text{A2})$$

The first term on the RHS equals zero. Thus:

$$P'(x) = \phi \int_0^{r_L} G(z) dz \quad (\text{A3})$$

gives us the discounted value of the *Put Option*.

Next, the discounted marginal value of the call option is  $C'(x) = \phi \int_{\sigma_H}^{\infty} (z - r_H) dF(\sigma)$ . With a change of variables:

$$C'(x) = \phi \int_{r_H}^{\infty} (z - r_H) dG(z) \quad (\text{A4})$$

Integrating by parts (using  $u = z - r_H$ ,  $du = dz$ ,  $dv = dG(z)$ , and  $v = G(z) - 1$ ), we get:

$$P'(x) = \phi \left( (z - r_H)(G(z) - 1) \Big|_{r_H}^{\infty} - \int_{r_H}^{\infty} (G(z) - 1) dz \right) \quad (\text{A5})$$

Again, the first term on the RHS equals zero. Thus:

$$C'(x) = \phi \int_{r_H}^{\infty} (1 - G(z)) dz \quad (\text{A6})$$

gives us the discounted value of the *Call Option*.

A mean-preserving spread in the distribution of  $\sigma$  causes both the left and the right tail of  $G(z)$  to become larger. This increases the integrated values of (A3) and (A6). The discounted values of both the call and put options increase. The former reduces the incentive to file patent applications while the latter increases it. Thus, the net effect on patent filings of greater uncertainty is ambiguous, as stated in the text.

## B Appendix: Patent Data Description

*PatentsView* is one of the main sources of patent data for this study.<sup>43</sup> It provides patent application level data such as patent numbers, application numbers, patent grant/issue date, application filing date, assignee, and technology category/classification for all public pre-grant applications and issued patents. It is an open data platform supported by the *Office of the Chief Economist at the U.S. Patent and Trademark office (USPTO)* that offers free access to raw and value-added data based on USPTO's publicly released bulk data.<sup>44</sup>

### B.1 Assignee Disambiguation

Identifying assignees, or owners, of patent applications is a challenging task. The USPTO does not assign a unique identifier to assignees at the time of application or grant and there are no requirements that impose a consistent format on the text field containing an assignee's name. To overcome this limitation, *PatentsView* employs a disambiguation algorithm to associate the same assignee name across different patent filings. Although this process does not use corporate structure to connect filings, *PatentsView's* disambiguation is a critical input for determining a total count of patent applications filed by firms (corporate entities) as described in Section 4.1.<sup>45</sup>

### B.2 Limitations

While *PatentsView* does provide data on the pre-grant publications of patent applications at the USPTO, the assignee of the patent application may be unrecorded at this time. Many assignment records are recorded just prior to the granting of a patent and these assignment records are typically more accurate than those previously recorded in the application process. Therefore, this study relies on the assignee information on granted patent applications whenever possible.

### B.3 Data Description

Compiling the granted patent component of our dependent variable (total application filings by firm  $i$  in quarter  $t$ ), involves linking the following files/tables on granted patent applications from *PatentsView*:<sup>46</sup>

- g\_application - provides the patent number, application number, patent type, and application filing date

---

<sup>43</sup><https://patentsview.org/>

<sup>44</sup>For more about this process see: <https://patentsview.org/data-pipeline>.

<sup>45</sup>See <https://patentsview.org/disambiguation>

<sup>46</sup>See <https://patentsview.org/download/data-download-tables>

- `g_assignee_disambiguated` – provides the disambiguated name of the organization of the assignee
- `g_cpc_current` – provides the current *CPC* symbols for each patent number
- `g_patent` – provides the patent number and grant date

The data on patents are first compiled using the `g_application` file in *PatentsView*.<sup>47</sup> The records in this file are filtered to include only utility patents (using the “`patent_application_type`” field), which are the focus of this study. This file is then linked to the `g_patent` file using the patent number (“`patent_id`” in *PatentsView*) as the key to join the data. In order to link the `g_assignee` file, the patent number is also used as the key. Using these tables, we are able to create our dataset on granted patents, which includes the assignee, patent number, application number, application filing date, and patent grant date for each utility patent in *PatentsView*.

Data on each firm’s technology field involve the use of the `g_cpc_current` file from *PatentsView*. The `g_cpc_current` file provides the CPC class symbols (“`cpc_class`” in *PatentsView*) and category (inventional vs additional) for each patent number. Only inventional *CPC* codes are included in our dataset, since they are reflective of the technology underlying the patent.

## B.4 Abandoned Patent Applications

To gather information on abandoned patent applications, both public and non-public, we use the USPTO’s Patent Application Locating and Monitoring (PALM) system.<sup>48</sup> The PALM system is the USPTO’s primary, internal system for handling and tracking patent applications. It includes both published and unpublished applications, allowing for a broader sample of abandoned applications available in *PatentsView*. Assignee data as well as patent family data associated with abandoned applications are provided by PALM and employed in the disambiguation process described below.

## B.5 Abandoned Application Assignee Disambiguation

Approximately one-third of abandoned applications have no assignment record available at the USPTO. Therefore, a different disambiguation methodology is necessary to identify the assignees of abandoned applications than utilized for granted patent applications.

In the absence of an assignee for an abandoned patent application, any applicant(s) provided in the PALM system are assumed to have ownership of the application and are treated as the assignee

---

<sup>47</sup>The *PatentsView* data dictionary provides details on data labels used in each file. See <https://patentsview.org/download/data-download-dictionary>

<sup>48</sup>Recall that abandoned applications are the second component of our dependent variable.

in the disambiguation process. This is consistent with USPTO’s rules for patent application processing and allows us to link a broader sample of patent applications to firms in our dataset.<sup>49</sup>

To begin the disambiguation process for abandoned applications, we first compare assignees with those listed on granted patent applications. If the assignee/applicant of an abandoned application exactly matches a disambiguated assignee on a granted patent application, or if there is a probabilistic match based on the Levenshtein distance with a threshold of 90%, and if both the abandoned application and the granted patent application are filed in the same year, then the assignee of the abandoned application is inferred to be the same assignee as the granted patent application.

Additionally, patent family history from PALM is utilized to infer the disambiguated assignee of abandoned applications.<sup>50</sup> If an abandoned application belongs to the same patent family as a granted patent application with a disambiguated assignee in our dataset, then it is assumed that the abandoned application belongs to the same assignee as the granted patent application, regardless of the year of application filing.

Finally, if there is a probabilistic match based on the Levenshtein distance with a threshold of 80% between all the inventors listed in an abandoned patent application and all the inventors listed on a disambiguated granted patent application, then it is assumed both applications belong to the same assignee regardless of the year of application.

---

<sup>49</sup>USPTO’s Manual of Patent Examining Procedures, Section 605. See <https://www.uspto.gov/web/offices/pac/mpep/s605.html>

<sup>50</sup>For our purposes, we define a patent family based on the priority of the patent application. For more detail, see: <https://www.uspto.gov/web/offices/pac/mpep/s210.html>.

## C Appendix: Sold Packages Data

A detailed dataset was provided by *Richardson Oliver Insights* (ROI) under arrangements that preserve confidential data. ROI is a leading provider of secondary patent market data. The following table provides an overview of the sample we analyzed:

| Variable                                   | Number (or Share/Value) |
|--------------------------------------------|-------------------------|
| Sale dates                                 | 2009 - 2021             |
| Packages                                   | 1,487                   |
| Patents (across packages)                  | 56,904                  |
| Technologies ( <i>CPC</i> class symbols)   | 116                     |
| Countries/offices:                         | 44                      |
| <i>Share, U.S.-issued patents</i>          | <i>75.0%</i>            |
| <i>Share, International-issued patents</i> | <i>25.0%</i>            |

In the sample, the average number of patents in a package was about 46. The median number of patents in a package was 8. The unit of trade was the package in that the asking prices pertained to a given package. Some packages are associated with exact prices, while others offered a range. In the latter case, the ROI data estimate the asking price to be a midpoint of the range.

## D Appendix: Data Sources

| Variable                                            | Description                                                                                                                                                                         | Data Source                                                                        |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Uncertainty ( $UNC$ )<br>(main measure)             | Market uncertainty as measured by the coefficient of variation of a firm's sales.                                                                                                   | Author calculations from S&P Capital IQ                                            |
| Uncertainty ( $UNC^A$ )<br>(autoregressive measure) | Market uncertainty as measured by the standard deviation of the residuals of an AR4 model of sales and time trend.                                                                  | Author calculations from S&P Capital IQ                                            |
| Uncertainty ( $UNC^s$ )<br>(stock market measure)   | Market uncertainty as measured by a firm's stock return relative to the market.                                                                                                     | Author calculations from S&P Capital IQ                                            |
| Uncertainty ( $UNC^B$ )<br>(business shock measure) | Firm revenues decomposed into cyclical and trend components.                                                                                                                        | Author calculations from S&P Capital IQ                                            |
| Patent applications                                 | The sum of granted patents and abandoned applications filed by a firm dated at the time of filing.                                                                                  | Author calculations from <i>PatentsView</i> Patents tables and USPTO's PALM system |
| Pre-sample patents                                  | The mean of patent applications during 1990-1998.                                                                                                                                   | Author calculations from patent applications derived                               |
| R&D Spending                                        | Expenses for the development of a new product or process, or the upgrading of existing product or service lines.                                                                    | S&P Capital IQ                                                                     |
| Sales (Revenue)                                     | Revenues earned from primary activities.                                                                                                                                            | S&P Capital IQ                                                                     |
| Gross Profit                                        | Total revenues minus the cost of revenues.                                                                                                                                          | S&P Capital IQ                                                                     |
| Full-time Employees                                 | The number of full-time employees.                                                                                                                                                  | S&P Capital IQ                                                                     |
| Cash Flow                                           | Cash from operations.                                                                                                                                                               | S&P Capital IQ                                                                     |
| Industry Concentration (HHI)                        | The Herfindahl-Hirschman Index of a firm's share of sales in a NAICS industry. HHI is mapped to lie in the interval $[0,1]$ .                                                       | Author calculations from S&P Capital IQ                                            |
| Index of Secondary Market ( $ISE$ )                 | An index measuring a firm's exposure to secondary markets for patents based on technology fields ( $CPC$ symbols). The fields are value weighted ( $v$ ) or count weighted ( $c$ ). | Author calculations from Richardson Oliver Insights                                |
| Credit Rating                                       | PAYDEX Score from Duns and Bradstreet for measuring firm credit risk.                                                                                                               | National Establishment Time-Series (NETS)<br><i>Wall &amp; Associates</i>          |

Figure 1: Increase in Put Value Option

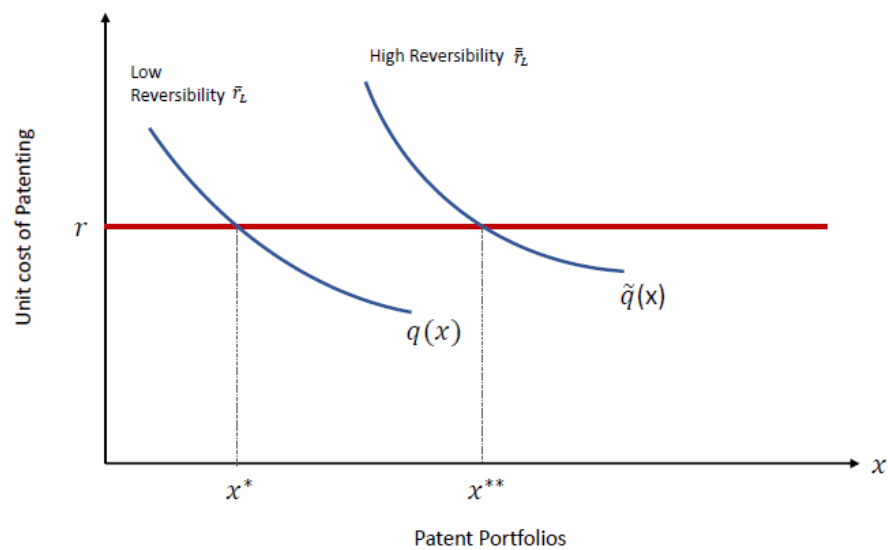


Figure 2: Patent Application Filings by Filing Year

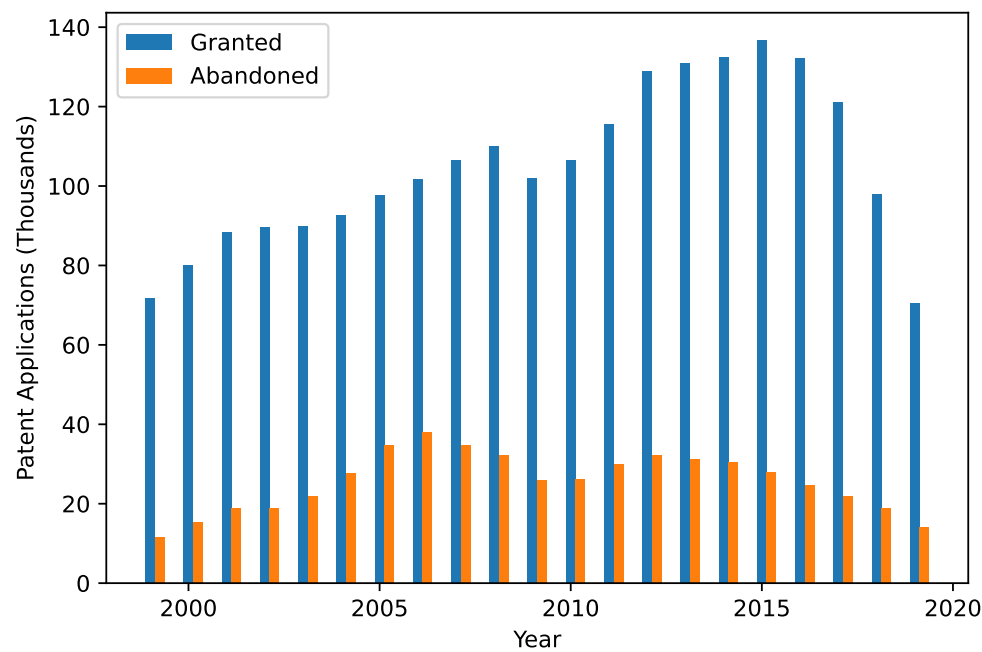
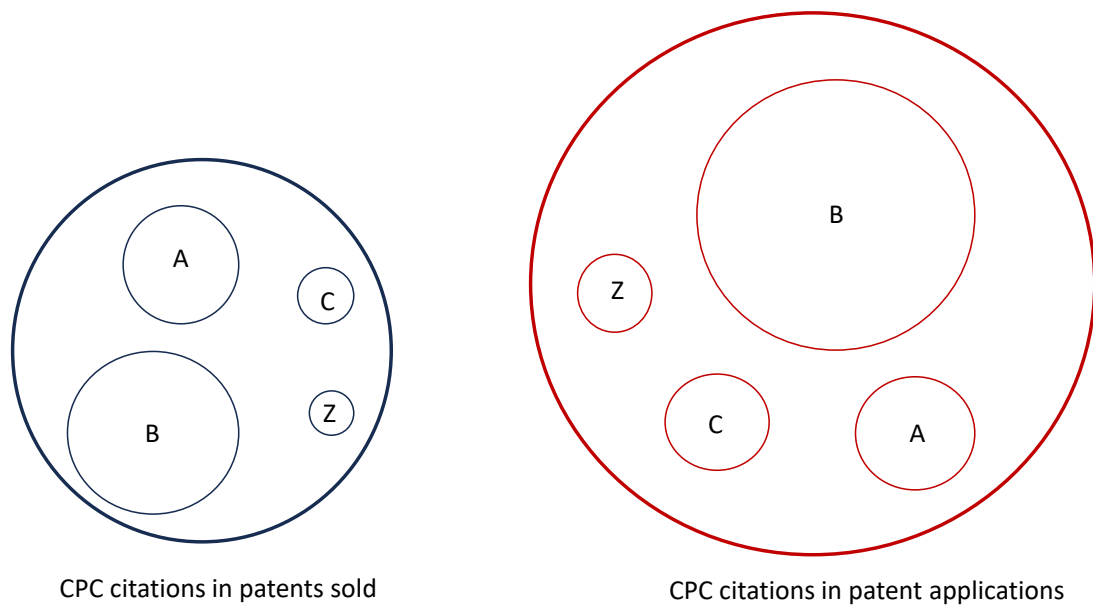


Figure 3: Shares of Technological Fields in Secondary Market and in Patent Filings



Notes: Each large circle represents the universe of technological fields cited in patents sold or in patent applications. The smaller circles inside represent the share of each technological field  $A$ ,  $B$ , ...,  $Z$  represented in patents sold or in patents applied for.

Figure 4: Index of Secondary Market Sales Exposure

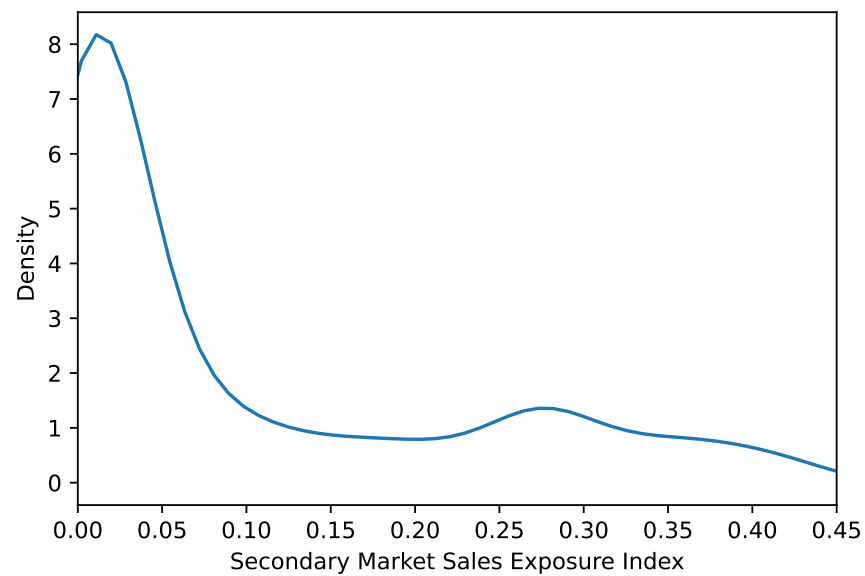


Figure 5: Patent Growth and Uncertainty, by Secondary Market Exposure

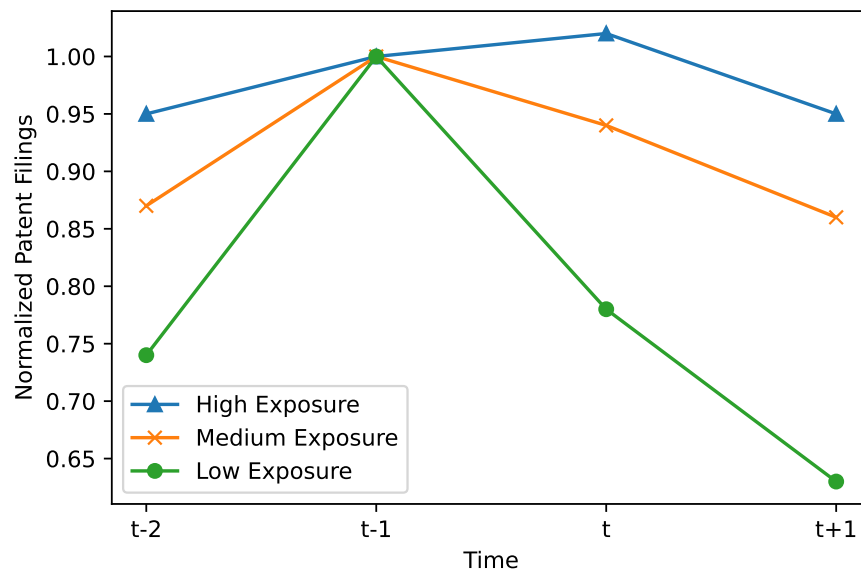


Table 1: Top 12 *CPC* classes in the secondary patent market and licensing market

| Secondary Patent Market Sales                |                  |               | U.S. Licensing Transactions             |                  |               |
|----------------------------------------------|------------------|---------------|-----------------------------------------|------------------|---------------|
| <i>CPC</i> Class Title                       | <i>CPC</i> Class | % share total | <i>CPC</i> Class Title                  | <i>CPC</i> Class | % share total |
| <b>Electric Communication Technique</b>      | H04              | 43.29         | <b>Medical or Veterinary Science</b>    | A61              | 18.09         |
| <b>Computing; Calculating; Counting</b>      | G06              | 27.20         | <b>Electric Communication Technique</b> | H04              | 17.56         |
| <b>Basic Electric Elements</b>               | H01              | 6.62          | <b>Computing; Calculating; Counting</b> | G06              | 15.24         |
| <b>Measuring; Testing</b>                    | G01              | 2.98          | <b>Basic Electric Elements</b>          | H01              | 6.80          |
| Basic Electronic Circuitry                   | H03              | 2.67          | Organic Chemistry                       | C07              | 5.41          |
| <b>Information Storage</b>                   | G11              | 2.07          | Checking Devices                        | G07              | 3.99          |
| Optics                                       | G02              | 1.69          | Physical or Chemical Processes          | B01              | 2.43          |
| Signaling                                    | G08              | 1.67          | <b>Measuring; Testing</b>               | G01              | 2.17          |
| <b>Medical or Veterinary Science</b>         | A61              | 1.52          | Engineering Elements                    | F16              | 1.58          |
| Musical Instruments; Acoustics               | G10              | 1.50          | Vehicles in General                     | B60              | 1.58          |
| Electric Techniques not elsewhere classified | H05              | 1.01          | <b>Information Storage</b>              | G11              | 1.44          |
| <b>Electric Power</b>                        | H02              | 0.90          | <b>Electric Power</b>                   | H02              | 1.42          |

In bold are *CPC* classes that match in each market. Time period: 2009-2021 for secondary patent markets and 2000-2021 for licensing. Data include all inventional *CPC* symbols for each patent.

Table 2: Sample Indexes of Secondary Market Sales, by Industry

| NAICS<br>code | Industry                                          | Index of<br>Sales Exposure |
|---------------|---------------------------------------------------|----------------------------|
| 5182          | Telecomm Data Process. Services                   | 0.3007                     |
| 5112          | Software publishers                               | 0.2602                     |
| 3342          | Communications equipment manuf.                   | 0.2062                     |
| 3343          | Audio and video equipment manuf.                  | 0.1962                     |
| 3341          | Computer and peripheral equipment manuf.          | 0.1959                     |
| 3344          | Semiconductors and other elect. comp. manuf.      | 0.1209                     |
| 5324          | Commercial and Industrial Machinery ...           | 0.0720                     |
| 2211          | Electric Power Gen., Trans, & Distr.              | 0.0481                     |
| 3364          | Aerospace products and parts manuf.               | 0.0347                     |
| 3332          | Industrial machinery                              | 0.0340                     |
| 3272          | Glass and glass products                          | 0.0338                     |
| 3391          | Medical equipment and supplies manuf.             | 0.0264                     |
| 3361          | Motor vehicles manufacturing                      | 0.0261                     |
| 3336          | Engines, turbines, and power trans. equip. manuf. | 0.0199                     |
| 3254          | Pharmaceuticals and Medicines                     | 0.0153                     |
| 3262          | Rubber Products                                   | 0.0108                     |
| 3261          | Plastic Products                                  | 0.0070                     |
| 3311          | Iron and Steel Mills and Ferro alloys             | 0.0067                     |
| 3251          | Basic Chemicals                                   | 0.0059                     |
| 6216          | Home Health Care Services                         | 0.0010                     |

Table 3: Descriptive Statistics

| Variable               | Mean   | Std<br>Deviation | Coef. of<br>Variation |
|------------------------|--------|------------------|-----------------------|
| Patent Applications    | 33.05  | 127.0            | 3.842                 |
| $UNC$                  | 0.230  | 0.460            | 2.000                 |
| $UNC^A$                | 0.130  | 0.188            | 1.446                 |
| $UNC^S$                | 2.316  | 2.381            | 1.028                 |
| $UNC^B$                | 0.142  | 0.868            | 6.105                 |
| $ISE^{m,v}$            | 0.075  | 0.107            | 1.420                 |
| $ISE^{m,c}$            | 0.068  | 0.093            | 1.356                 |
| $ISE^{g,v}$            | 3.878  | 3.850            | 0.993                 |
| $ISE^{g,c}$            | 0.040  | 0.043            | 1.071                 |
| R&D Spending           | 67.8   | 295.9            | 4.364                 |
| Sales                  | 2,575  | 7,767            | 3.016                 |
| Cash Flow              | 356.0  | 1,354            | 3.803                 |
| Credit Rating          | 71.15  | 5.50             | 0.077                 |
| Full-time Employees    | 28,372 | 98,257           | 3.463                 |
| Industry Concentration | 0.286  | 0.243            | 8.490                 |
| Pre-Sample Patents     | 12.93  | 48.18            | 3.726                 |

Authors' calculations. Data are quarterly.

Operating data are in millions of real (2012) dollars.

Patent Applications and Full-time Employees are in counts.

Variable definitions are in Appendix D.

Table 4: Main Results

|                                              | Total Patent Applications |                      |                      |                      |                      |                      |
|----------------------------------------------|---------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                              | (1)                       | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
| <i>UNC</i>                                   | -2.074***<br>(0.227)      | -2.556***<br>(0.263) | -2.841***<br>(0.265) | -1.655***<br>(0.257) | -2.005***<br>(0.293) | -2.879***<br>(0.341) |
| <i>ISE<sup>m,v</sup></i>                     |                           | 5.244***<br>(0.344)  | 4.877***<br>(0.426)  |                      | 2.707***<br>(0.295)  | 1.699***<br>(0.348)  |
| <i>UNC</i> $\times$ <i>ISE<sup>m,v</sup></i> |                           |                      | 1.877*<br>(1.057)    |                      |                      | 5.507***<br>(0.914)  |
| R&D Stock                                    |                           |                      |                      | 0.265***<br>(0.015)  | 0.236***<br>(0.014)  | 0.238***<br>(0.013)  |
| Sales                                        |                           |                      |                      | 0.020***<br>(0.006)  | 0.018***<br>(0.006)  | 0.017***<br>(0.006)  |
| Cash Flow                                    |                           |                      |                      | 0.066***<br>(0.016)  | 0.061***<br>(0.016)  | 0.062***<br>(0.016)  |
| Credit Rating                                |                           |                      |                      | 0.010*<br>(0.005)    | 0.004<br>(0.005)     | 0.004<br>(0.005)     |
| Full-time Employees                          |                           |                      |                      | 0.057<br>(0.350)     | 0.137<br>(0.348)     | 0.186<br>(0.344)     |
| Industry Concentration                       |                           |                      |                      | -0.651***<br>(0.135) | -0.725***<br>(0.134) | -0.806***<br>(0.137) |
| Pre-Sample Mean                              | 0.012***<br>(0.000)       | 0.011***<br>(0.000)  | 0.011***<br>(0.000)  | 0.008***<br>(0.000)  | 0.008***<br>(0.000)  | 0.008***<br>(0.000)  |
| Constant                                     | 2.462***<br>(0.168)       | 2.502***<br>(0.190)  | 2.540***<br>(0.191)  | 0.857**<br>(0.420)   | 1.503***<br>(0.408)  | 1.616***<br>(0.414)  |
| Observations                                 | 35,112                    | 35,112               | 35,112               | 35,112               | 35,112               | 35,112               |
| Pseudo $R^2$                                 | 0.580                     | 0.618                | 0.618                | 0.707                | 0.716                | 0.718                |
| Log Likelihood                               | -817,831.5                | -744,318.3           | -743,817.7           | -570,455.8           | -552,335.8           | -548,846.2           |
| LR test statistic                            |                           | 147,026.5***         | 1,001.1***           |                      | 36,240.1***          | 6,979.0***           |

Newey-West (autocorrelation and heteroskedasticity-consistent) standard errors in parentheses.

Likelihood ratio (LR) test statistic tests the successive addition of *ISE* and *UNC*  $\times$  *ISE*.

\*\*\* p<0.01, \*\* p<0.05, \* p<0.1. Period and industry fixed effects included.

Table 5: **Alternative Indexes of Secondary Market Exposure**

|                                              | Total Patent Applications |                      |                      |                      |                      |                      |
|----------------------------------------------|---------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                                              | (1)                       | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  |
| <i>UNC</i>                                   | -2.009***<br>(0.293)      | -3.082***<br>(0.323) | -2.044***<br>(0.298) | -5.381***<br>(0.443) | -2.065***<br>(0.298) | -4.829***<br>(0.397) |
| <i>ISE<sup>m,c</sup></i>                     | 3.051***<br>(0.329)       | 1.728***<br>(0.140)  |                      |                      |                      |                      |
| <i>UNC</i> $\times$ <i>ISE<sup>m,c</sup></i> |                           | 7.232***<br>(1.033)  |                      |                      |                      |                      |
| <i>ISE<sup>g,c</sup></i>                     |                           |                      | 7.147***<br>(0.571)  | 3.684***<br>(0.653)  |                      |                      |
| <i>UNC</i> $\times$ <i>ISE<sup>g,c</sup></i> |                           |                      |                      | 23.928***<br>(2.279) |                      |                      |
| <i>ISE<sup>g,v</sup></i>                     |                           |                      |                      |                      | 0.091***<br>(0.008)  | 0.047***<br>(0.009)  |
| <i>UNC</i> $\times$ <i>ISE<sup>g,v</sup></i> |                           |                      |                      |                      |                      | 0.288***<br>(0.036)  |
| R&D Stock                                    | 0.239***<br>(0.014)       | 0.241***<br>(0.013)  | 0.218***<br>(0.013)  | 0.214***<br>(0.013)  | 0.213***<br>(0.013)  | 0.210***<br>(0.013)  |
| Sales                                        | 0.018***<br>(0.006)       | 0.018***<br>(0.006)  | 0.019***<br>(0.006)  | 0.019***<br>(0.005)  | 0.018***<br>(0.006)  | 0.018***<br>(0.005)  |
| Cash Flow                                    | 0.061***<br>(0.016)       | 0.062***<br>(0.016)  | 0.054***<br>(0.016)  | 0.052***<br>(0.016)  | 0.053***<br>(0.016)  | 0.052***<br>(0.016)  |
| Credit Rating                                | 0.004<br>(0.005)          | 0.004<br>(0.005)     | -0.000<br>(0.005)    | -0.002<br>(0.005)    | -0.000<br>(0.005)    | -0.001<br>(0.005)    |
| Full-time Employees                          | 0.135<br>(0.349)          | 0.165<br>(0.344)     | 0.108<br>(0.347)     | -0.000<br>(0.337)    | 0.158<br>(0.343)     | 0.136<br>(0.334)     |
| Industry Concentration                       | -0.701***<br>(0.133)      | -0.800***<br>(0.135) | -0.563***<br>(0.126) | -0.875***<br>(0.129) | -0.619***<br>(0.127) | -0.898***<br>(0.129) |
| Pre-Sample Mean                              | 0.008***<br>(0.000)       | 0.008***<br>(0.000)  | 0.007***<br>(0.000)  | 0.007***<br>(0.000)  | 0.007***<br>(0.000)  | 0.007***<br>(0.000)  |
| Constant                                     | 1.473***<br>(0.404)       | 1.631***<br>(0.401)  | 1.787***<br>(0.400)  | 2.469***<br>(0.396)  | 1.890***<br>(0.413)  | 2.404***<br>(0.414)  |
| Observations                                 | 35,112                    | 35,112               | 35,028               | 35,028               | 35,028               | 35,028               |
| Pseudo $R^2$                                 | 0.717                     | 0.719                | 0.726                | 0.735                | 0.725                | 0.732                |
| Log Likelihood                               | -551,614.4                | -546,897.8           | -532,192.1           | -515,865.4           | -535,549.7           | -522,180.5           |

Newey-West standard errors in parentheses.

*ISE<sup>m,c</sup>* is based on simple counts of secondary market transactions.*ISE<sup>g,c</sup>* is based on the ratio of secondary market transactions to patent applications. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1.

Period and industry fixed effects included.

Table 6: Alternative Measures of Uncertainty

|                          | Total Patent Applications |                      |                      |                      |                      |                      |                      |                      |                      |
|--------------------------|---------------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|----------------------|
|                          | (1)                       | (2)                  | (3)                  | (4)                  | (5)                  | (6)                  | (7)                  | (8)                  | (9)                  |
| $UNC^A$                  | -0.008***<br>(0.002)      | -0.010***<br>(0.002) | -0.017***<br>(0.003) |                      |                      |                      |                      |                      |                      |
| $UNC^S$                  |                           |                      |                      | -0.019*<br>(0.011)   | -0.005**<br>(0.389)  | -0.050***<br>(0.014) |                      |                      |                      |
| $UNC^B$                  |                           |                      |                      |                      |                      |                      | -0.589***<br>(0.205) | -0.602***<br>(0.213) | -1.210***<br>(0.405) |
| $ISE^{m,v}$              |                           | 2.012***<br>(0.290)  | 1.479***<br>(0.298)  |                      | 1.732***<br>(0.293)  | 1.226***<br>(0.351)  |                      | 1.763***<br>(0.290)  | 1.496***<br>(0.326)  |
| $UNC^A \times ISE^{m,v}$ |                           |                      | 0.044***<br>(0.015)  |                      |                      |                      |                      |                      |                      |
| $UNC^S \times ISE^{m,v}$ |                           |                      |                      |                      |                      | 0.386***<br>(0.072)  |                      |                      |                      |
| $UNC^B \times ISE^{m,v}$ |                           |                      |                      |                      |                      |                      |                      |                      | 3.795**<br>(1.704)   |
| R&D Stock                | 0.298***<br>(0.017)       | 0.279***<br>(0.016)  | 0.276***<br>(0.016)  | 0.284***<br>(0.017)  | 0.268***<br>(0.016)  | 0.269***<br>(0.016)  | 0.285***<br>(0.016)  | 0.267***<br>(0.016)  | 0.265***<br>(0.015)  |
| Sales                    | 0.024***<br>(0.007)       | 0.023***<br>(0.007)  | 0.022***<br>(0.007)  | 0.022***<br>(0.006)  | 0.022***<br>(0.006)  | 0.023***<br>(0.006)  | 0.022***<br>(0.006)  | 0.021***<br>(0.006)  | 0.021***<br>(0.006)  |
| Cash Flow                | 0.064***<br>(0.019)       | 0.061***<br>(0.019)  | 0.059***<br>(0.019)  | 0.060***<br>(0.018)  | 0.055***<br>(0.018)  | 0.052***<br>(0.018)  | 0.060***<br>(0.018)  | 0.055***<br>(0.018)  | 0.056***<br>(0.017)  |
| Credit Rating            | 0.014***<br>(0.005)       | 0.010***<br>(0.005)  | 0.009***<br>(0.005)  | 0.014***<br>(0.005)  | 0.010***<br>(0.005)  | 0.009***<br>(0.005)  | 0.014***<br>(0.005)  | 0.010***<br>(0.005)  | 0.009***<br>(0.005)  |
| Full-time Employees      | 0.216<br>(0.407)          | 0.287<br>(0.415)     | 0.334<br>(0.412)     | 0.414<br>(0.398)     | 0.473<br>(0.409)     | 0.372<br>(0.406)     | 0.370<br>(0.390)     | 0.454<br>(0.401)     | 0.457<br>(0.399)     |
| Industry Concentration   | -0.527***<br>(0.148)      | -0.595***<br>(0.151) | -0.603***<br>(0.150) | -0.521***<br>(0.144) | -0.524***<br>(0.146) | -0.461***<br>(0.144) | -0.577***<br>(0.149) | -0.609***<br>(0.151) | -0.619***<br>(0.153) |
| Pre-sample Mean          | 0.007***<br>(0.000)       | 0.007***<br>(0.000)  | 0.007***<br>(0.000)  | 0.008***<br>(0.000)  | 0.007***<br>(0.000)  | 0.007***<br>(0.000)  | 0.008***<br>(0.000)  | 0.008***<br>(0.000)  | 0.008***<br>(0.000)  |
| Constant                 | -1.724***<br>(0.503)      | -1.192***<br>(0.495) | 1.029**<br>(0.499)   | 0.501<br>(0.394)     | 0.835**<br>(0.389)   | 0.972**<br>(0.390)   | 0.421<br>(0.404)     | 0.804**<br>(0.398)   | 0.888**<br>(0.395)   |
| Observations             | 32,186                    | 32,186               | 32,186               | 35,112               | 35,112               | 35,112               | 35,112               | 35,112               | 35,112               |
| Pseudo $R^2$             | 0.691                     | 0.697                | 0.697                | 0.689                | 0.693                | 0.696                | 0.691                | 0.695                | 0.695                |
| Log Likelihood           | -56400.4                  | -553902.7            | -552807.2            | -605120.8            | -597314.8            | -593536.4            | -602577.1            | -594315.2            | -593060.9            |

Newey-West standard errors in parentheses.  $UNC^S$  is based on stock price volatility,  $UNC^A$  on AR residuals, and  $UNC^B$  on business cycle volatility. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1. Period and industry fixed effects included.

Table 7: Sector Breakdown

| Industry Group         | <i>Coefficient Estimates:</i> |             |                        |                        | Implied Threshold | Number of Observ. | Number of Firms | # Firms above Threshold (%) | Median $ISE^{m,v}$ |
|------------------------|-------------------------------|-------------|------------------------|------------------------|-------------------|-------------------|-----------------|-----------------------------|--------------------|
|                        | $UNC$                         | $ISE^{m,v}$ | $UNC \times ISE^{m,v}$ | $UNC \times ISE^{m,v}$ |                   |                   |                 |                             |                    |
| Capital Goods          | -3.635***                     | 11.55***    | 4.587                  | 4.587                  | not signif        | 7,224             | 86              | n/a                         | 0.012              |
| Consumer Durables      | -1.231***                     | -2.557***   | 5.165**                | 5.165**                | 0.238             | 2,940             | 35              | 3 (8.6%)                    | 0.007              |
| Food, Bev, Tobacco     | -1.498*                       | -7.742      | -35.73                 | -35.73                 | not signif        | 1,260             | 15              | n/a                         | 0.007              |
| Information Technology | -0.352*                       | -2.029***   | 2.792***               | 2.792***               | 0.126             | 8,400             | 100             | 48 (48.0%)                  | 0.121              |
| Materials              | -1.029*                       | 12.41***    | 90.34**                | 90.34**                | 0.011             | 3,780             | 45              | 14 (31.1%)                  | 0.005              |
| Pharmaceuticals        | -0.450***                     | 3.853       | -7.222                 | -7.222                 | not signif        | 2,856             | 34              | n/a                         | 0.011              |
| Retail & Marketing     | -18.37***                     | -1.788      | 57.53***               | 57.53***               | 0.320             | 1,680             | 22              | 2 (9.1%)                    | 0.081              |
| Services               | -2.019***                     | -0.817      | 1.548                  | 1.548                  | not signif        | 3,108             | 37              | n/a                         | 0.031              |
| Transportation         | -6.701***                     | 13.70*      | 34.19                  | 34.19                  | not signif        | 1,428             | 17              | n/a                         | 0.010              |
| Utilities & Energy     | -1.372**                      | 43.90***    | 11.61**                | 11.61**                | 0.118             | 2,268             | 27              | 10 (37.0%)                  | 0.008              |
| OVERALL                |                               |             |                        |                        |                   | 35,112            | 418             | 77 (18.4%)                  | 0.018              |

Industry groups are based on S&P Capital IQ Primary industries. Info. technology includes hardware, software, and semiconductors. Pharmaceuticals includes biotechnology. Materials includes chemicals. Capital goods include machinery. Services cover financial, commercial, and consumer services. Consumer durables include household products. Transportation covers automobiles, rail, and air. Retail and marketing include internet commerce and distribution. \*\*\* p<0.01, \*\* p<0.05, \* p<0.1. n/a denotes not applicable.

Table 8: **Magnitude of Uncertainty on Patent Application Filings**

| Group           | Mean Filings | Mean $ISE^{m,v}$ | Mean Uncertainty | Elasticity Estimate | % Change in Uncert. | % Change in Filings |
|-----------------|--------------|------------------|------------------|---------------------|---------------------|---------------------|
| Pooled          | 26           | 0.070            | 0.230            | -0.290              | 23.3                | -3%                 |
| Low Exposure    | 3            | 0.003            | 0.172            | -0.378              | 21.1                | -8.2%               |
| Medium Exposure | 22           | 0.026            | 0.251            | -0.369              | 22.8                | -4.8%               |
| High Exposure   | 58           | 0.221            | 0.246            | -0.026              | 26.3                | -1.3%               |

| <i>By Select Firms:</i>        |              |                  |                  |                     |                     |            |
|--------------------------------|--------------|------------------|------------------|---------------------|---------------------|------------|
| Low Exposure Group             | Mean Filings | Mean $ISE^{m,v}$ | Mean Uncertainty | Elasticity Estimate | % Change in Uncert. | $\Delta P$ |
| BorgWarner Inc.                | 18           | 0.003            | 0.111            | -0.74               | 24.6                | -3.34      |
| Harley-Davidson, Inc.          | 2            | 0.003            | 0.102            | -0.68               | 28.3                | -0.44      |
| Eastman Chemical Company       | 15           | 0.004            | 0.105            | -0.07               | 26.4                | -0.28      |
| Sealed Air Corporation         | 13           | 0.004            | 0.077            | -0.05               | 41.3                | -0.27      |
| American Axle & Manuf Holdings | 5            | 0.004            | 0.181            | -1.21               | 31.6                | -1.93      |
| Goodyear Tire & Rubber Co.     | 22           | 0.004            | 0.028            | -0.19               | 23.2                | -0.96      |
| PPG Industries, Inc.           | 20           | 0.005            | 0.056            | -0.03               | 22.4                | -0.15      |
| Chevron Corporation            | 36           | 0.005            | 0.161            | -0.21               | 47.4                | -3.62      |

| Medium Exposure Group         | Mean Filings | Mean $ISE^{m,v}$ | Mean Uncertainty | Elasticity Estimate | % Change in Uncert. | $\Delta P$ |
|-------------------------------|--------------|------------------|------------------|---------------------|---------------------|------------|
| Pfizer Inc.                   | 70           | 0.009            | 0.137            | -0.06               | 22.7                | -0.98      |
| Procter & Gamble Company      | 157          | 0.009            | 0.028            | -0.03               | 33.5                | -1.73      |
| Merck & Co., Inc.             | 70           | 0.010            | 0.239            | -0.11               | 30.4                | -2.28      |
| Raytheon Technologies Corp.   | 157          | 0.018            | 0.087            | -0.32               | 17.5                | -8.69      |
| 3M Company                    | 168          | 0.025            | 0.055            | -0.94               | 17.5                | -27.60     |
| Thermo Fisher Scientific Inc. | 42           | 0.028            | 0.257            | -0.12               | 33.0                | -1.60      |
| Danaher Corporation           | 68           | 0.050            | 0.171            | -0.62               | 18.1                | -7.67      |
| Boeing Company                | 210          | 0.056            | 0.087            | -0.32               | 14.0                | -9.36      |
| Gentex Corporation            | 10           | 0.061            | 0.210            | -1.41               | 18.2                | -2.54      |
| Lockheed Martin Corporation   | 98           | 0.063            | 0.092            | -0.33               | 20.0                | -6.53      |

| High Exposure Group         | Mean Filings | Mean $ISE^{m,v}$ | Mean Uncertainty | Elasticity Estimate | % Change in Uncert. | $\Delta P$ |
|-----------------------------|--------------|------------------|------------------|---------------------|---------------------|------------|
| Motorola Solutions, Inc.    | 221          | 0.175            | 1.116            | 0.15                | 63.6                | 21.42      |
| IBM Corp.                   | 1659         | 0.197            | 0.019            | -0.04               | 50.3                | -31.27     |
| Apple Inc.                  | 303          | 0.247            | 0.283            | 0.10                | 23.3                | 6.73       |
| NVIDIA Corporation          | 53           | 0.251            | 0.287            | 0.10                | 31.7                | 1.68       |
| Amazon.com, Inc.            | 198          | 0.280            | 0.095            | -0.22               | 20.2                | -8.63      |
| Microsoft Corporation       | 635          | 0.299            | 0.041            | 0.02                | 9.8                 | 1.24       |
| Qualcomm Incorporated       | 409          | 0.314            | 0.227            | 0.12                | 12.5                | 6.11       |
| Verizon Communications Inc. | 104          | 0.381            | 0.084            | -0.17               | 32.7                | -5.77      |
| Juniper Networks, Inc.      | 41           | 0.390            | 0.399            | 0.29                | 19.5                | 2.34       |
| Cisco Systems, Inc.         | 209          | 0.392            | 0.177            | 0.13                | 34.0                | 9.36       |

The means of variables are quarterly values.  $\Delta P$  denotes the change in patent applications upon a  $\frac{1}{2}$  **standard deviation increase in uncertainty**. From Equation (9), the elasticity is computed as:  $\epsilon = \frac{\partial P}{\partial UNC} \frac{UNC}{P} = (\hat{\beta}_1 + \hat{\beta}_2 ISE) \times UNC$ .